

MATH5311 notes
ADVANCED NUMERICAL METHOD (I)

LOL

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Chapter 1

Stability analysis, local truncation error analysis, convergence analysis

1.1 Truncation error

Notes 1.1.1: Grid and Taylor approximation

We introduce grid in space and time:

$$\Delta x = \frac{1}{N}, x_j = j\Delta x, \Delta t = \frac{T}{M}, t_n = n\Delta t$$

Then we have Taylor expansion of u

$$u(x_{i+1}) = u(x_i) + \Delta x u_x(x_i) + \frac{1}{2} \Delta x^2 u_{xx}(x_i) + \dots$$

Definition 1.1.1: Big-O

$f = O(g)$ if there exists a constant C such that

$$|f| \leq C|g|$$

Example 1.1.1:

Approximation of u_x :

$$\begin{aligned} u_x &\approx \frac{u(x_{i+1}) - u(x_i)}{\Delta x} = D_{+x}u(x_i) \text{ (Forward difference)} \\ &\approx \frac{u(x_i) - u(x_{i-1}))}{\Delta x} = D_{-x}u(x_i) \text{ (Backward difference)} \\ &\approx \frac{u(x_{i+1}) - u(x_{i-1}))}{2\Delta x} = D_0u(x_i) \text{ (Central difference)} \\ &\approx \frac{u(x_{i+1/2}) - u(x_{i-1/2}))}{\Delta x} = \delta_x u(x_i) \text{ (Central difference 2)} \end{aligned}$$

Notes 1.1.2:

We can use Taylor expansion around x_i to analyse the accuracy of difference approximation

Example 1.1.2:*Central difference:*

$$\begin{aligned}
u(x_{i+1}) &= u(x_i) + \Delta x u_x(x_i) + \frac{1}{2} \Delta x^2 u_{xx}(x_i) + \frac{1}{6} \Delta x^3 u_{xxx}(x_i) + O(\Delta x^4) \\
u(x_{i-1}) &= u(x_i) - \Delta x u_x(x_i) + \frac{1}{2} \Delta x^2 u_{xx}(x_i) - \frac{1}{6} \Delta x^3 u_{xxx}(x_i) + O(\Delta x^4) \\
\Rightarrow \frac{u(x_{i+1}) - u(x_{i-1}))}{2\Delta x} &= u_x(x_i) + \frac{1}{3} \Delta x^2 u_{xxx}(x_i) \\
&= u_x(x_i) + O(\Delta x^2)
\end{aligned}$$

Second order derivative:

$$\begin{aligned}
u(x_{i+1}) &= u(x_i) + \Delta x u_x(x_i) + \frac{1}{2} \Delta x^2 u_{xx}(x_i) + \frac{1}{6} \Delta x^3 u_{xxx}(x_i) + \frac{1}{24} \Delta x^4 u_{xxxx}(x_i) + O(\Delta x^5) \\
u(x_i) &= u(x_i) \\
u(x_{i-1}) &= u(x_i) - \Delta x u_x(x_i) + \frac{1}{2} \Delta x^2 u_{xx}(x_i) - \frac{1}{6} \Delta x^3 u_{xxx}(x_i) + \frac{1}{24} \Delta x^4 u_{xxxx}(x_i) + O(\Delta x^5) \\
\Rightarrow \frac{u(x_{i+1}) - 2u(x_i) + u(x_{i-1}))}{\Delta x^2} &= u_{xx}(x_i) + \frac{1}{12} \Delta x^2 u_{xxxx}(x_i) + O(\Delta x^3) \\
&= u_{xx}(x_i) + O(\Delta x^2)
\end{aligned}$$

Notes 1.1.3:*We can also use Taylor expansion around x_i to get the approximation***Example 1.1.3:***We let*

$$u_{xx}(x_i) \approx Au(x_{i+1}) + Bu(x_i) + Cu(x_{i-1}))$$

Then by Taylor we have

$$(A + B + C)u(x_i) + \Delta x(A - C)u_x(x_i) + \frac{1}{2}\Delta x^2(A + C)u_{xx}(x_i) + \frac{1}{6}\Delta x^3(A - C)u_{xxx}(x_i) + \dots$$

By comparing we have

$$\begin{cases} u(x_i) : & A + B + C = 0 \\ u_x(x_i) : & \Delta x(A - C) = 0 \\ u_{xx}(x_i) : & \frac{1}{2}\Delta x^2(A + C) = 1 \end{cases} \Rightarrow A = C = \frac{1}{\Delta x^2}, B = \frac{-2}{\Delta x^2}$$

Definition 1.1.2: Truncation error*If $L_h u_j^n = G_h$ is a scheme approximation to PDE $Lu = G$, and $u(x, t)$ is the solution to this PDE, then*

$$T(x, t) = L_h u(x, t) - G_h$$

If the local truncation error

$$T_j^n := T(x_j, t_n) = O(\Delta t^p) + O(\Delta x^q)$$

*Then we say the numerical scheme is p th order in time and q th order in space***Notes 1.1.4: Truncation error***Truncation error means the error by replacing differential operator(e.g. $\partial_t, \partial_{xx}$) with finite difference approximation*

Example 1.1.4: Truncation error

We consider

$$u_t = u_{xx}$$

Let $u(x_j, t_n) = u_j^n$ then

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

Now if $u_t = u_{xx}$, then local truncation error is

$$\begin{aligned} T_j^n &= \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} - \frac{u(x_{j+1}^n - 2u(x_j^n) + u(x_{j-1}^n))}{\Delta x^2} \\ &= u_t(x_j, t_n) - u_{xx}(x_j, t_n) + \Delta t \frac{1}{2} u_{tt}(x_j, t_n) - \frac{4x^2}{12} u_{xxxx}(x_j, t_n) + O(\Delta t^2) + O(\Delta x^4) \\ &= O(\Delta t) + O(\Delta x^2) \end{aligned}$$

as u is exact solution to $u_t = u_{xx}$

1.2 Implementaion and stability**Example 1.2.1: Implementaion**

We consider $u_t = u_{xx}$ with boundary condition

$$BC: u(0, t) = u(1, t) = 0$$

$$IC: u(x, 0) = f(x)$$

Then by letting $r = \frac{\Delta t}{\Delta x^2}$ we have the scheme

$$u_j^{n+1} = (1 - 2r)u_j^n + ru_{j+1}^n + ru_{j-1}^n \quad j = 1, \dots, N-1$$

For $j = 0$ and $j = N$, by BC we have

$$u_0^n = u_N^n = 0$$

And initial

$$u_j^0 = f(x_j) \quad j = 0, \dots, N$$

Hence we can Implement as following:

STEP 1: Initialization, $u_j^0 = f(x_j) \quad j = 0, \dots, N$

STEP 2 Update with $u_j^{n+1} = (1 - 2r)u_j^n + ru_{j+1}^n + ru_{j-1}^n \quad j = 1, \dots, N-1$

Impose BC: $u_0^{n+1} = 0, u_N^{n+1} = 0$

STEP 3: $n = n + 1$ and back to step 2, until reach $T = n\Delta t$

Definition 1.2.1: Consistency and convergence

For truncation error T_j^n

We say our method is consistent if

$$T_j^n \rightarrow 0 \text{ as } \Delta t \rightarrow 0, \Delta x \rightarrow 0$$

We say our method is convergent if the numerical solution u_j^n satisfy

$$u_j^n - u(x_j, t_n) \rightarrow 0 \text{ as } \Delta t \rightarrow 0, \Delta x \rightarrow 0$$

Notes 1.2.1: Consistency and convergence

Consistency $\neq$ Convergence

Consistency means one step error due to differential operator with FD approximation

Meanwhile convergence require error accumulate in a controllable manner

Example 1.2.2:

We consider the scheme

$$u_j^{n+1} = (1 - 2r)u_j^n + ru_{j+1}^n + ru_{j-1}^n$$

We define error

$$e_j^n = u(x_j, t_n) - u_j^n$$

By definition of T_j^n :

$$T_j^n = \frac{u(x_j, t_{n+1}) - u(x_j, t_n)}{\Delta t} - \frac{u(x_{j+1}^n - 2u(x_j^n) + u(x_{j-1}^n))}{\Delta x^2}$$

Rearrange we have

$$u(x_j, t_{n+1}) = u(x_j, t_n) + r(u(x_{j+1}, t_n) - 2u(x_j, t_n) + u(x_{j-1}, t_n)) + \Delta t T_j^n$$

Hence we have

$$\begin{aligned} e_j^{n+1} &= e_j^n + r(e_{j+1}^n - 2e_j^n + e_{j-1}^n) + \Delta t T_j^n \\ &= (1 - 2r)e_j^n + r(e_{j+1}^n + e_{j-1}^n) + \Delta t T_j^n \end{aligned}$$

We let

$$E^n = \max_j \{|e_j^n|\}, T^n = \max_j \{|T_j^n|\}$$

If $0 \leq r \leq \frac{1}{2} \Rightarrow 1 - 2r > 0$

Hence we have

$$|e_j^{n+1}| \leq (1 - 2r)|e_j^n| + r|e_{j+1}^n| + r|e_{j-1}^n| + \Delta t |T_j^n|$$

Now as BC is exact we have $e_0^{n+1} = e_N^{n+1} = 0$

And IC is exact we have $E^0 = 0$

And

$$|T^i| \leq A(\Delta t + \Delta x^2)$$

as $|T^i| \in O(\Delta t + \Delta x^2)$

Therefore

$$\begin{aligned} |E^{n+1}| &\leq (1 - 2r)|E^n| + r|E^n| + r|E^n| + \Delta t |T^n| \\ &= |E^n| + \Delta t |T^n| \\ &\leq |E^{n-1}| + \Delta t |T^{n-1}| + \Delta t |T^n| \\ &\leq \dots \\ &\leq |E^0| + \Delta t \sum_{i=1}^n |T^i| \\ &= \Delta t \sum_{i=1}^n |T^i| \\ &\leq \Delta t n A(\Delta t + \Delta x^2) \\ &= t_n A(\Delta t + \Delta x^2) \\ &\leq T A(\Delta t + \Delta x^2) \xrightarrow{\Delta t + \Delta x^2 \rightarrow 0} 0 \end{aligned}$$

If $r > \frac{1}{2}$ then $1 - 2r < 0$

Therefore

$$\begin{aligned} |E^{n+1}| &\leq -(1 - 2r)|E^n| + r|E^n| + r|E^n| + \Delta t |T^n| \\ &= (4r - 1)|E^n| + \Delta t |T^n| \\ &\leq \Delta t \sum_{i=1}^n (4r - 1)^{n-i} T^i \end{aligned}$$

Which is unclear how to control

Notes 1.2.2: Fourier analysis

We put

$$u_j^n = \lambda^n e^{ikx_j}$$

Then we put into the scheme and find the range of $|\lambda|$

We say the scheme is stable if there exists K_T such that

$$|\lambda(k)|^n \leq K_T \quad \forall n\Delta t \leq T, \forall k$$

Von-Neumann condition:

$$|\lambda| \leq 1 + \alpha\Delta t$$

In most case we just look at condition $|\lambda(k)| \leq 1$

Stable: $|\lambda| \leq 1$ for all modes

Unconditionally stable: Stable for all $\Delta t > 0$

Unstable: find k to make $|\lambda| > 1$

Example 1.2.3: Fourier analysis

Again the scheme

$$u_j^{n+1} = (1 - 2r)u_j^n + ru_{j+1}^n + ru_{j-1}^n$$

Then put $u_j^n = \lambda^n e^{ikx_j}$

$$\lambda^{n+1} e^{ikx_j} = (1 - 2r)\lambda^n e^{ikx_j} + r\lambda^n e^{ik(x_j + \Delta x)} + r\lambda^n e^{ik(x_j - \Delta x)}$$

$$\lambda = (1 - 2r) + re^{ik\Delta x} + re^{-ik\Delta x}$$

$$= 1 - 2r(1 - \cos(k\Delta x))$$

$$= 1 - 4r \sin^2 \frac{k\Delta x}{2}$$

Now

$$|\lambda| \leq 1 \iff -1 \leq 1 - 4r \sin^2 \frac{k\Delta x}{2} \leq 1$$

$$\iff r \leq \frac{2}{4 \sin^2 \frac{k\Delta x}{2}}$$

If $r \leq \frac{1}{2}$, then it is true as $0 \leq \sin^2 \frac{k\Delta x}{2} \leq 1$

If $r > \frac{1}{2}$, then put $k\Delta x = \pi$, then

$$\lambda = 1 - 4r \sin^2 \frac{\pi}{2} < -1$$

1.3 Stability and implicit methods**Notes 1.3.1: Implicit (Backward Euler in time)**

We have

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2}$$

Put $r = \frac{\Delta t}{\Delta x^2}$ then

$$(1 + 2r)u_j^{n+1} - ru_{j+1}^{n+1} - ru_{j-1}^{n+1} = u_j^n$$

Truncation error = $O(\Delta t) + O(\Delta x^2)$

Fourier:

$$\lambda = \frac{1}{1 + 4r \sin^2 \frac{k\Delta x}{2}}$$

Hence for any $k\Delta x$ we have $|\lambda| \leq 1$
Hence the method is Unconditionally stable
Implementaion:

$$(1 + 2r)u_j^{n+1} - ru_{j+1}^{n+1} - ru_{j-1}^n = u_j^n \quad 1 \leq j \leq N - 1$$

Then

$$U^n = \begin{pmatrix} u_1^n \\ \vdots \\ u_{N-1}^n \end{pmatrix}$$

Then we can rewrite as linear system

$$\begin{pmatrix} 1+2r & -r & & & \\ -r & 1+2r & -r & & \\ & \ddots & \ddots & \ddots & \\ & & -r & 1+2r & r \\ & & & -r & 1+2r \end{pmatrix} \begin{pmatrix} u_1^{n+1} \\ u_2^{n+1} \\ \vdots \\ u_{N-2}^{n+1} \\ u_{N-1}^{n+1} \end{pmatrix} = \begin{pmatrix} u_1^n \\ u_2^n \\ \vdots \\ u_{N-2}^n \\ u_{N-1}^n \end{pmatrix}$$

Then solve $AU^{n+1} = U^n$

Notes 1.3.2: Crank-Nicolson

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \frac{1}{2} \left(\frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2} + \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2} \right)$$

If

$$\delta_x^2 u_j = \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2}$$

Then

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \delta_x^2 \left(\frac{u_j^{n+1} + u_j^n}{2} \right)$$

Fourier analysis: $u_j^n = \lambda^n e^{ikx_j}$

$$\begin{aligned} \delta_x^2 e^{ikx_j} &= \frac{1}{\Delta x^2} (e^{ikx_{j+1}} - 2e^{ikx_j} + e^{-ikx_{j-1}}) \\ &= \frac{1}{\Delta x^2} (e^{ik\Delta x} - 2 + e^{-ik\Delta x}) e^{ikx_j} \\ &= \frac{1}{\Delta x^2} (2\cos(k\Delta x) - 2) e^{ikx_j} \\ &= \frac{-4}{\Delta x^2} \sin^2 \frac{k\Delta x}{2} e^{ikx_j} \end{aligned}$$

$$u_j^{n+1} = u_j^n + \frac{1}{2} \Delta t \delta_x^2 u_j^n + \frac{1}{2} \Delta t \delta_x^2 u_j^{n+1}$$

$$\left(1 - \frac{1}{2} \Delta t \delta_x^2\right) u_j^{n+1} = \left(1 + \frac{1}{2} \Delta t \delta_x^2\right) u_j^n$$

$$\left(1 + 2r \sin^2 \frac{k\Delta x}{2}\right) \lambda^{n+1} e^{ikx_j} = \left(1 - 2r \sin^2 \frac{k\Delta x}{2}\right) \lambda^n e^{ikx_j}$$

$$\lambda = \frac{1 - 2r \sin^2 \frac{k\Delta x}{2}}{1 + 2r \sin^2 \frac{k\Delta x}{2}}$$

$$\left|1 - 2r \sin^2 \frac{k\Delta x}{2}\right| \leq |1| + \left|2r \sin^2 \frac{k\Delta x}{2}\right|$$

$$= 1 + 2r \sin^2 \frac{k\Delta x}{2}$$

$$\Rightarrow |\lambda| \leq 1 \text{ for any } r > 0$$

Hence Unconditionally stable

Truncation error: $O(\Delta t^2) + O(\Delta x^2)$

Implementaion:

$$\begin{pmatrix} 1+r & -\frac{1}{2}r & & & \\ -\frac{1}{2}r & 1+r & -\frac{1}{2}r & & \\ & \ddots & \ddots & \ddots & \\ & & -\frac{1}{2}r & 1+r & -\frac{1}{2}r \\ & & & -\frac{1}{2}r & 1+r \end{pmatrix} \begin{pmatrix} u_1^{n+1} \\ u_2^{n+1} \\ \vdots \\ u_{N-2}^{n+1} \\ u_{N-1}^{n+1} \end{pmatrix} = \begin{pmatrix} 1-r & \frac{1}{2}r & & & \\ \frac{1}{2}r & 1-r & \frac{1}{2}r & & \\ & \ddots & \ddots & \ddots & \\ & & \frac{1}{2}r & 1-r & \frac{1}{2}r \\ & & & \frac{1}{2}r & 1-r \end{pmatrix} \begin{pmatrix} u_1^n \\ u_2^n \\ \vdots \\ u_{N-2}^n \\ u_{N-1}^n \end{pmatrix}$$

If we let

$$L = \begin{pmatrix} -r & \frac{1}{2}r & & & \\ \frac{1}{2}r & -r & \frac{1}{2}r & & \\ & \ddots & \ddots & \ddots & \\ & & \frac{1}{2}r & -r & \frac{1}{2}r \\ & & & \frac{1}{2}r & -r \end{pmatrix}$$

Then

$$(I - L)U^{n+1} = (I + L)U^n + BC$$

Notes 1.3.3: θ -scheme

θ -scheme:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = \theta \frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2} + (1 - \theta) \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

When

- $\theta > 0$: a linear solver is needed
- $\theta = 0$: Explicit scheme(Forward Euler)
- $\theta = 1/2$: Crank Nicolson
- $\theta = 1$: Implicit scheme(Backward Euler)

Notes 1.3.4:

If $u_h^{n+1} = Q_h u_h^n + G_h$ is the scheme solving the PDE $Lu = G$

Then Truncation error:

$$\Delta t T_h^n = u(x_j, t_{n+1}) - Q_h u(x_j, t_n) - G_h$$

Stability:

$$\|Q_h^n u_h\| \leq K_T \|u_h\| \quad \forall n \Delta t \leq T$$

Theorem 1.3.1: Lax-Theorem

$u_h^{n+1} = Q_h u_h^n + G_h$ is consistent with p th order accuracy in space and q th order accuracy in time, Q_h linear operator

And this scheme is stable wrt $\|\cdot\|_h$

Then it converge to exact solution as $\Delta x, \Delta t \rightarrow 0$ if initial error $\|e_n^0\| \rightarrow 0$

Proof.

$$u_h^{n+1} = Q_h u_h^n + G_h$$

$$e_h^{n+1} = (u_0^n - u(x_0, t_n), \dots, u_j^n - u(x_j, t_n))$$

$$\text{Consistency: } u(x, t_{n+1}) = Q_h u(x, t_n) + G_h + \Delta t T_h^n$$

$$e_h^{n+1} = Q_h e_h^n - \Delta t T_h^n$$

$$= Q_h^2 (e_h^{n-1} - \Delta t T_h^{n-1}) - \Delta t T_h^n$$

$$= \dots$$

$$= Q_h^{n+1} e_h^0 - \Delta t \sum_{m=0}^n Q_h^{n-m} T_h^m$$

$$\begin{aligned} \text{Stability: } \|Q_h^{n+1} e_h^0\| &\leq K_T \|e_h^0\| \\ \|Q_h^{n+1} T_h^m\| &\leq K_T \|T_h^m\| \leq K_T (O(\Delta x^p) + O(\Delta t^q)) \\ \Rightarrow \|e_h^{n+1}\| &\leq K_T \|e_h^0\| + (n+1) \Delta t K_T (O(\Delta x^p) + O(\Delta t^q)) \\ &= K_T (\|e_h^0\| + T(O(\Delta x^p) + O(\Delta t^q))) \xrightarrow{\Delta x, \Delta t \rightarrow 0} 0 \end{aligned}$$

□

Notes 1.3.5:

The error accumulation is independent of G_h

For stability analysis, in most case it is good enough to look at homogeneous problem without source and zero boundary condition

1.4 Time stepping and general BCs**Notes 1.4.1: Method of line for time stepping**

STEP 1: Keep time continuous and apply spatial discretization, and we obtain system of ODE
 $u_t = F(u, t)$

STEP 2: Apply ODE solver to solve

$$u_t = F(u, t)$$

(1) Forward Euler:

$$u^{n+1} = u^n + \Delta t F(u^n, t_n)$$

(2) Backward Euler:

$$u^{n+1} = u^n + \Delta t F(u^{n+1}, t_{n+1})$$

(3) Implicit midpoint:

$$u^{n+1} = u^n + \Delta t F\left(\frac{u^n + u^{n+1}}{2}, t_{n+1/2}\right)$$

(4) Runge-Kutta method:

2nd order ($O(\Delta t^2)$ error):

$$\begin{aligned} u^{n+1/2} &= u^n + \frac{\Delta t}{2} F(u^n, t_n) \\ u^{n+1} &= u^n + \Delta t F(u^{n+1/2}, t_{n+1/2}) \end{aligned}$$

4th order ($O(\Delta t^4)$ error):

$$\begin{aligned} k_1 &= \Delta t F(u^n, t_n) \\ k_2 &= \Delta t F\left(u^n + \frac{1}{2} k_1, t_n + \frac{1}{2} \Delta t\right) \\ k_3 &= \Delta t F\left(u^n + \frac{1}{2} k_2, t_n + \frac{1}{2} \Delta t\right) \\ k_4 &= \Delta t F(u^n + k_3, t_n + \Delta t) \\ u^{n+1} &= u^n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \end{aligned}$$

(5) Multi-step method:

$$u^{n+1} = u^n + \frac{3}{2} \Delta t F(u^n, t_n) - \frac{1}{2} \Delta t F(u^{n-1}, t_n)$$

Notes 1.4.2: General BC

STEP 1: Use finite difference approximation and ghost points to approximate BC

STEP 2L Write out equation for update of boundary points

STPE 3: Write out matrix-vector form

Example 1.4.1: Dirichlet Boundary Condition

Dirichlet Boundary Condition:

$$\begin{aligned}u_t &= u_{xx} \\ u(0, t) &= \alpha(t) \\ u(1, t) &= \beta(t)\end{aligned}$$

Interior points:

$$u_j^{n+1} = (1 - 2r)u_j^n + ru_{j+1}^n + ru_{j-1}^n \quad j = 1, \dots, N - 1$$

Then

$$u_0^n = \alpha(t_n), \quad u_N^n = \beta(t_n)$$

For $j = 1$:

$$\begin{aligned}u_1^{n+1} &= (1 - 2r)u_1^n + ru_2^n + ru_0^n \\ &= (1 - 2r)u_1^n + ru_2^n + r\alpha(t_n)\end{aligned}$$

For $j = N - 1$:

$$u_{N-1}^{n+1} = (1 - 2r)u_{N-1}^n + ru_{N-2}^n + r\beta(t_n)$$

Matrix-vector form:

$$\begin{pmatrix} u_1^{n+1} \\ u_2^{n+1} \\ \vdots \\ u_{N-2}^{n+1} \\ u_{N-1}^{n+1} \end{pmatrix} = \begin{pmatrix} 1-2r & r & & & \\ r & 1-2r & r & & \\ & \ddots & \ddots & \ddots & \\ & & r & 1-2r & r \\ & & & r & 1-2r \end{pmatrix} \begin{pmatrix} u_1^n \\ u_2^n \\ \vdots \\ u_{N-2}^n \\ u_{N-1}^n \end{pmatrix} + \begin{pmatrix} r\alpha(t_n) \\ 0 \\ \vdots \\ 0 \\ r\beta(t_n) \end{pmatrix}$$

Example 1.4.2: Neumann Boundary Condition

Neumann Boundary Condition:

$$\begin{aligned}u_t &= u_{xx} \\ u_x(0, t) &= \alpha(t) \\ u(1, t) &= \beta(t)\end{aligned}$$

Interior points:

$$u_j^{n+1} = (1 - 2r)u_j^n + ru_{j+1}^n + ru_{j-1}^n \quad j = 1, \dots, N - 1$$

Then

$$u_N^n = \beta(t_n)$$

For $j = N - 1$:

$$u_{N-1}^{n+1} = (1 - 2r)u_{N-1}^n + ru_{N-2}^n + r\beta(t_n)$$

And by central difference:

$$\begin{aligned}u_x(0, t) &= \alpha(t) \\ \frac{u_1^n - u_{-1}^n}{2\Delta x} &= \alpha(t_n) \\ u_{-1} &= u_1^n - 2\Delta x\alpha(t_n)\end{aligned}$$

For $j = 0$:

$$u_0^{n+1} = (1 - 2r)u_0^n + ru_1^n + ru_{-1}^n$$

$$= (1 - 2r)u_0^n + 2ru_1^n - 2r\Delta x\alpha(t_n)$$

Matrix-vector form: We have to start with $j = 0$

$$\begin{pmatrix} u_0^{n+1} \\ u_1^{n+1} \\ \vdots \\ u_{N-2}^{n+1} \\ u_{N-1}^{n+1} \end{pmatrix} = \begin{pmatrix} 1-2r & r & & & \\ r & 1-2r & r & & \\ & \ddots & \ddots & \ddots & \\ & & & r & 1-2r & r \\ & & & & r & 1-2r \end{pmatrix} \begin{pmatrix} u_0^n \\ u_1^n \\ \vdots \\ u_{N-2}^n \\ u_{N-1}^n \end{pmatrix} + \begin{pmatrix} -2r\Delta x\alpha(t_n) \\ 0 \\ \vdots \\ 0 \\ r\beta(t_n) \end{pmatrix}$$

Example 1.4.3: Robin boundary condition

Left BC:

$$u_x(0, t) = \alpha(t)u(0, t) + g(t)$$

Then

$$\begin{aligned} \frac{u_1 - u_{-1}}{2\Delta x} &= \alpha(t_n)u_0^n + g(t_n) \\ u_{-1} &= u_1^n - 2\Delta x\alpha(t_n)u_0^n - 2\Delta xg(t_n) \\ u(0, t_n)_{xx} &= \frac{u_1^n - 2u_0^n + u_{-1}}{\Delta x^2} \\ &= \frac{1}{\Delta x^2}(2u_1^n - 2(1 + \Delta x\alpha(t_n))u_0^n - 2\Delta xg(t_n)) \end{aligned}$$

We consider Crank-Nicolson

$$u_j^{n+1} - \frac{1}{2}r(u_{j-1}^{n+1} - 2u_j^{n+1} + u_{j+1}^{n+1}) = u_j^n + \frac{1}{2}r(u_{j-1}^n - 2u_j^n + u_{j+1}^n)$$

Put $j = 0$ then

$$\begin{aligned} u_0^{n+1} - \frac{1}{2}r(-2(1 + \Delta x\alpha(t_{n+1}))u_0^{n+1} + 2u_1^{n+1} - 2\Delta xg(t_{n+1})) \\ = u_0^n - \frac{1}{2}r(-2(1 + \Delta x\alpha(t_n))u_0^n + 2u_1^n - 2\Delta xg(t_n)) \\ \Rightarrow (1 + r(1 + \Delta x)\alpha(t_{n+1}))u_0^{n+1} - ru_1^{n+1} \\ = (1 + r(1 + \Delta x)\alpha(t_n))u_0^n - ru_1^n - r\Delta x(g(t_n) + g(t_{n+1})) \end{aligned}$$

Notes 1.4.3:

Fourier analysis may fail as scheme may not hold at $j = 0$

Can use energy stability analysis or normal mode analysis

Matrix may not be symmetric but we can divide in BC row to make it symmetric

Chapter 2

Finite difference method

2.1 Finite difference for 2D heat equation

Notes 2.1.1: 2D heat equation

2D heat equation:

$$u_t = \Delta u = u_{xx} + u_{yy}, \quad (x, y) \in [0, 1]^2$$

$$BC: u(x, y, t) = g(x, y, t), \quad (x, y) \in \partial\Omega$$

$$IC: u(x, y, 0) = f(x, y)$$

$$u(x_j, y_k, t_n) = u(j\Delta x, k\Delta y, n\Delta t), \quad \Delta x = \frac{1}{N_x}, \Delta y = \frac{1}{N_y}$$

$$u_{xx} : D_{+x}D_{-x}u_{jk}^n = \partial_x^2 u_{jk}^n = \frac{u_{j+1,k}^n - 2u_{jk}^n + u_{j-1,k}^n}{\Delta x^2}$$

$$u_{yy} : D_{+y}D_{-y}u_{jk}^n = \partial_y^2 u_{jk}^n = \frac{u_{j,k+1}^n - 2u_{jk}^n + u_{j,k-1}^n}{\Delta y^2}$$

Notes 2.1.2: Forward Euler

Interior points:

$$\frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} = (D_{+x}D_{-x}u_{jk}^n + D_{+y}D_{-y}u_{jk}^n)$$

Boundary:

$$u_{jk}^n - g(x_j, y_k, t_n) \text{ if } j = 0 \text{ or } k = 0 \text{ or } j = N_x \text{ or } k = N_y$$

Stability:

$$r_x + r_y = \frac{\Delta t}{\Delta x^2} + \frac{\Delta t}{\Delta y^2} \leq \frac{1}{2} \Rightarrow \Delta t \leq \frac{\Delta x^2 \Delta y^2}{\Delta x^2 + \Delta y^2} = O(h^2)$$

If $h \approx \Delta x$ and $h \approx \Delta y$

Algorithm:

STEP 1: Set up mesh and choose Δt based on stability

STEP 2: Initialize U^0 based on IC

STEP 3: Update U^{n+1} from U^n until final time

Notes 2.1.3: Crank Nicolson

Crank Nicolson:

$$\frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} = (D_{+x}D_{-x} + D_{+y}D_{-y}) \frac{u_{jk}^n + u_{jk}^{n+1}}{2}$$

Matrix Vector:

$$\left(I - \frac{1}{2}r_x L_x - \frac{1}{2}r_y L_y\right) U^{n+1} = \left(I + \frac{1}{2}r_x L_x + \frac{1}{2}r_y L_y\right) U^n$$

Where L_x, L_y correspond to $D_{+x}D_{-x}$ and $D_{+y}D_{-y}$ respectively

Stability:

$$u_{jl}^n = \lambda^n e^{i(k_x x_j + k_y y_l)}$$

$$\left(1 - \frac{\Delta t}{2}D_{+x}D_{-x} - \frac{\Delta t}{2}D_{+y}D_{-y}\right) u_{jl}^{n+1} = \left(1 + \frac{\Delta t}{2}D_{+x}D_{-x} + \frac{\Delta t}{2}D_{+y}D_{-y}\right) u_{jl}^n$$

Now

$$\begin{aligned} D_{+x}D_{-x}u_{jl} &= \frac{u_{j+1,l} - 2u_{jl} + u_{j-1,l}}{\Delta x^2} \\ &= \frac{1}{\Delta x^2}(e^{ik_x \Delta x} + e^{-ik_x \Delta x} - 2)e^{i(k_x x_j + k_y y_l)} \\ &= \frac{1}{\Delta x^2}(2\cos(k_x \Delta x) - 2)e^{i(k_x x_j + k_y y_l)} \\ &= \frac{-4}{\Delta x^2} \sin^2 \frac{k_x \Delta x}{2} e^{i(k_x x_j + k_y y_l)} \\ D_{+y}D_{-y} &= \frac{-4}{\Delta y^2} \sin^2 \frac{k_y \Delta y}{2} e^{i(k_x x_j + k_y y_l)} \end{aligned}$$

Therefore

$$\lambda = \frac{1 - 2r_x \sin^2 \frac{k_x \Delta x}{2} - 2r_y \sin^2 \frac{k_y \Delta y}{2}}{1 + 2r_x \sin^2 \frac{k_x \Delta x}{2} + 2r_y \sin^2 \frac{k_y \Delta y}{2}}$$

Hence $|\lambda| \leq 1$ and hence Unconditionally stable

In practice we take $\Delta t = \min(\Delta x, \Delta y)$ or $\sqrt{\Delta x^2 + \Delta y^2}$

Notes 2.1.4: ADI: alternating direction implicit algorithm

First half step: u_{xx} implicit

Second half step: u_{yy} implicit

$$\begin{aligned} \frac{u_{jk}^{n+1/2} - u_{jk}^n}{1/2\Delta t} &= D_{+x}D_{-x}u_{jk}^{n+1/2} + D_{+y}D_{-y}u_{jk}^n \\ \frac{u_{jk}^{n+1} - u_{jk}^{n+1/2}}{1/2\Delta t} &= D_{+x}D_{-x}u_{jk}^{n+1/2} + D_{+y}D_{-y}u_{jk}^{n+1} \end{aligned}$$

Fourier analysis:

$$\begin{aligned} \left(1 - \frac{\Delta t}{2}D_{+x}D_{-x}\right) u_{jk}^{n+1/2} &= \left(1 - \frac{\Delta t}{2}D_{+y}D_{-y}\right) u_{jk}^n \\ \left(1 - \frac{\Delta t}{2}D_{+y}D_{-y}\right) u_{jk}^{n+1} &= \left(1 - \frac{\Delta t}{2}D_{+x}D_{-x}\right) u_{jk}^{n+1/2} \end{aligned}$$

Hence we have

$$u_{jk}^{n+1} = \left(1 - \frac{\Delta t}{2}D_{+y}D_{-y}\right)^{-1} \left(1 + \frac{\Delta t}{2}D_{+x}D_{-x}\right) \left(1 - \frac{\Delta t}{2}D_{+x}D_{-x}\right)^{-1} \left(1 + \frac{\Delta t}{2}D_{+y}D_{-y}\right) u_{jk}^n$$

$$u_{jl}^n = \lambda^n e^{i(k_x x_j + k_y y_l)}$$

Then

$$\begin{aligned} D_{+x}D_{-x}e^{i(k_x x_j + k_y y_l)} &= -4 \sin^2 \frac{k_x \Delta x}{2} \frac{1}{\Delta x^2} \\ D_{+y}D_{-y}e^{i(k_x x_j + k_y y_l)} &= -4 \sin^2 \frac{k_y \Delta y}{2} \frac{1}{\Delta y^2} \end{aligned}$$

Hence

$$\lambda = \frac{(1 - 2r_x \sin^2 \frac{k_x \Delta x}{2})(1 - 2r_y \sin^2 \frac{k_y \Delta y}{2})}{(1 + 2r_x \sin^2 \frac{k_x \Delta x}{2})(1 + 2r_y \sin^2 \frac{k_y \Delta y}{2})}$$

Hence $|\lambda| \leq 1$ for all r_x, r_y , Unconditionally stable

Truncation error:

$$\begin{aligned} \frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} &= D_{+x} D_{-x} u_{jk}^{n+1/2} + D_{+y} D_{-y} \left(\frac{1}{2} (u_{jk}^{n+1} + u_{jk}^n) \right) \\ \frac{u_{jk}^{n+1} - u_{jk}^n - 2u_{jk}^{n+1/2}}{1/2 \Delta t} &= D_{+y} D_{-y} (u_{jk}^{n+1} + u_{jk}^n) \\ \frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} &= (D_{+x} D_{-x} + D_{+y} D_{-y}) \left(\frac{1}{2} (u_{jk}^{n+1} + u_{jk}^n) \right) \\ &\quad - \frac{\Delta t^2}{4} D_{+x} D_{-x} D_{+y} D_{-y} \left(\frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} \right) \\ \left(1 + \frac{\Delta t^2}{4} D_{+x} D_{-x} D_{+y} D_{-y} \right) \left(\frac{u_{jk}^{n+1} - u_{jk}^n}{\Delta t} \right) &= (D_{+x} D_{-x} + D_{+y} D_{-y}) \left(\frac{1}{2} (u_{jk}^{n+1} + u_{jk}^n) \right) \end{aligned}$$

Put exact solution then $T = O(\Delta t^2 + \Delta x^2 + \Delta y^2)$

Implementaion:

First half:

$$\begin{pmatrix} I - \frac{\Delta t}{2} L_x & & & \\ & I - \frac{\Delta t}{2} L_x & & \\ & & \ddots & \\ & & & I - \frac{\Delta t}{2} L_x \end{pmatrix} \begin{pmatrix} U_1^{(x),n+1/2} \\ U_2^{(x),n+1/2} \\ \vdots \\ U_{N_y+1}^{(x),n+1/2} \end{pmatrix} = b^n$$

Second half:

$$\begin{pmatrix} I - \frac{\Delta t}{2} L_y & & & \\ & I - \frac{\Delta t}{2} L_y & & \\ & & \ddots & \\ & & & I - \frac{\Delta t}{2} L_y \end{pmatrix} \begin{pmatrix} U_1^{(y),n+1/2} \\ U_2^{(y),n+1/2} \\ \vdots \\ U_{N_x+1}^{(y),n+1/2} \end{pmatrix} = b^{n+1/2}$$

Notes 2.1.5: Cost of solving linear system

$\mathbb{R}^{N \times N} : N = O(N_x N_y)$

Crank-Nicolson

- LU: $O(N^3) = O(N_x^3 N_y^3)$
- Conjugate Gradient (CG): $O(N^2) = O(N_x^2 N_y^2)$
- Multigrid (MG): $O(N_x N_y \log(N_x N_y))$
- Thomas Algorithm: NOT work (only work for tridiagonal matrices)

ADI: First half: Solve $\mathbb{R}^{N_x \times N_x}$ equation $O(N_y)$ times

Second half: Solve $\mathbb{R}^{N_y \times N_y}$ equation $O(N_x)$ times

- LU: $O(N^3) = O(N_x N_y (N_x^2 + N_y^2))$
- Conjugate Gradient (CG): $O(N^2) = O(N_x N_y (N_x + N_y))$
- Multigrid (MG): $O(N_x \log(N_x)) N_y + O(N_y \log(N_y)) N_x$
- Thomas Algorithm: $O(N_x) N_y + O(N_y) N_x$

2.2 Finite difference for 1d advection

Notes 2.2.1: Continuous model problem

The problem

$$\begin{aligned} u_t + cu_x &= 0 \\ u(x, 0) &= u_0(x) \end{aligned}$$

Exact solution is $u(x, t) = u_0(x - ct)$

Notes 2.2.2: Numerical scheme

We try the following

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = 0$$

Test the stability $u_j^n = \lambda^n e^{ikx_j}$ then

$$\begin{aligned} \lambda &= 1 - i\sigma \sin(k\Delta x) \\ |\lambda| &= \sqrt{1 + \sigma^2 \sin^2(k\Delta x)} \geq 1 \end{aligned}$$

Where $\sigma = \frac{c\Delta t}{\Delta x}$ Hence it is Unconditionally Unstable

And we can see that the solution propagate from left to right when $c > 0$

Notes 2.2.3: Upwind discretization

Upwind:

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_j^n - u_{j-1}^n}{\Delta x} = 0 \quad c > 0$$

Then $u_j^n = \lambda^n e^{ikx_j}$, $\sigma = \frac{c\Delta t}{\Delta x}$ then

$$\lambda = (1 - \sigma + \sigma \cos(k\Delta x)) - i\sigma \sin(k\Delta x)$$

Then

$$|\lambda| \leq 1 \iff 2(1 - \sigma)\sigma(\cos(k\Delta x) - 1) \leq 0 \text{ holds for any } k\Delta x$$

We need

$$2\sigma(1 - \sigma) \geq 0$$

As

$$\sigma = \frac{c\Delta t}{\Delta x} > 0 \Rightarrow 1 - \sigma > 0 \Rightarrow c\Delta t \leq \Delta x$$

$$(cu_x)_j \approx \begin{cases} c_j \frac{u_j - u_{j-1}}{\Delta x} & c > 0 \\ c_j \frac{u_{j+1} - u_j}{\Delta x} & c < 0 \end{cases}$$

Accuracy: $O(\Delta t) + O(\Delta x)$

Stability: $c\Delta t \leq \Delta x$

Definition 2.2.1: CFL condition

The discrete domain of dependence for convergent scheme cover the continuous domain of dependence

Notes 2.2.4: CFL condition

- Scheme violate CFL is not convergent
- Scheme obey CFL may or may not convergent

For central: $\Delta t \leq \frac{1}{c} \Delta x$

Notes 2.2.5: Accuracy

We can obtain arbitrary order accuracy

STEP 1: Taylor expansion

STEP 2: Exploit PDE to convert time derivative to spatial

If IC is given by $u(0, t) = g(t)$, we convert spatial derivative to time derivative

Example 2.2.1: Lax-Wendroff

Second order accurate

Taylor:

$$u(x_j, t_{n+1}) = u(x_j, t_n) + \Delta t u_t(x_j, t_n) + \frac{1}{2} \Delta t^2 u_{tt}(x_j, t_n) + O(\Delta t^3)$$

Utilize PDE

$$u_t = -cu_x$$

$$u_{tt} = -cu_{xt} = -cu_{tx} = c^2 u_{xx}$$

Hence

$$u(x_j, t_{n+1}) = u(x_j, t_n) - c\Delta t u_x(x_j, t_n) + \frac{1}{2} \Delta t^2 c^2 u_{xx}(x_j, t_n) + O(\Delta t^3)$$

Use central difference for spatial derivative

$$u_j^{n+1} = u_j^n - c\Delta t \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} + \frac{1}{2} \Delta t^2 c^2 \frac{u_{j+1}^n - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

Stability: $\frac{c\Delta t}{\Delta x} \leq 1$

2.3 Dispersion analysis and modified equation**Notes 2.3.1: Wave speed and group velocity**

$u(x, t) = \cos(kx - wt) = \cos\left(k\left(x - \frac{w}{k}t\right)\right)$ with $\frac{w}{k} = \text{wave speed}$

Propagation of two waves with same amplitude

$$u(x, t) = \cos(\Delta t - \Delta kx) \cos(w_c t - k_c x)$$

Then group velocity/speed of envelope: $v_g = \frac{\Delta w}{\Delta k}$

$\Delta w \rightarrow 0, \Delta k \rightarrow 0, v_g = \frac{\partial w}{\partial k}$ group velocity

Notes 2.3.2: Dispersion analysis on continuous level

Put

$$u(x, t) = e^{i(kx - wt)}$$

Example 2.3.1: Dispersion analysis on continuous level

Consider

$$u_t + cu_x = 0$$

Substitute

$$-iwe^{i(kx-wt)} + ikce^{i(kx-wt)} = 0$$

$$w = kc$$

Therefore

$$u(x, t) = e^{i(kx-wt)} = e^{ik(x-ct)}$$

Notes 2.3.3: Dispersion analysis on discrete level

STEP 1: Taylor expansion

STEP 2: Keep leading error term in truncation error and drop high order ones

STEP 3: Convert time derivative to spatial ones using PDE ($w = f(k)$ to make algebra simpler)STEP 4: Substituting $e^{i(kx-wt)}$ into modified equation

If it make the amplitude then the scheme is dissipative

If it change the phase speed then the scheme is dispersive

Example 2.3.2: Dispersion analysis on discrete level

(1) Upwind scheme

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_j^n - u_{j-1}^n}{\Delta x} = 0$$

Modified equation:

$$u(x, t + \Delta t) = u(x, t) + u_t \Delta t + \frac{1}{2} \Delta t^2 u_{tt} + O(\Delta t^3)$$

$$u(x - \Delta x, t) = u(x, t) - u_x \Delta x + \frac{1}{2} \Delta x^2 u_{xx} + O(\Delta x^3)$$

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} = u_t + \frac{1}{2} \Delta t u_{tt} + O(\Delta t^2)$$

$$\frac{u_j^n - u_{j-1}^n}{\Delta x} = u_x - \frac{1}{2} \Delta x u_{xx} + O(\Delta x^2)$$

Then

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_j^n - u_{j-1}^n}{\Delta x} = u_t + cu_x + \frac{1}{2} \Delta t u_{tt} - c \frac{1}{2} \Delta x u_{xx} + O(\Delta t^2)$$

Put $u_t = -cu_x$, $u_{tt} = c^2 u_{xx}$ then

$$u_t + cu_x = \frac{1}{2} c \Delta x (1 - \sigma) u_{xx}$$

is the modified equation

Put

$$u = e^{i(kx-wt)}$$

Then

$$w = ck - \frac{i}{2} c \Delta x (1 - \sigma) k^2$$

Hence

$$u = e^{\frac{-1}{2} c \Delta x (1 - \sigma) k^2 t} e^{i(k(x-ct))}$$

Hence dissipative as it make amplitude decay

(2) Lax-Wendroff

$$\frac{u_j^{n+1} - u_j^n}{\Delta t} + c \frac{u_{j+1}^n - u_{j-1}^n}{2\Delta x} = \frac{1}{2} c^3 \Delta t \frac{u_j^{n+1} - 2u_j^n + u_{j-1}^n}{\Delta x^2}$$

Now by Taylor

$$\begin{aligned} u_t + cu_x &= \frac{-1}{6}\Delta t^2 u_{ttt} - \frac{1}{2}\Delta t u_{tt} - \frac{1}{6}c\Delta x^2 u_{xxx} + \frac{1}{2}c^2\Delta t u_{xx} \\ &= \frac{-1}{6}c\Delta x^2(1 - \sigma^2)u_{xxx} \end{aligned}$$

$u = e^{i(kx - wt)}$ then

$$w = ck + \frac{1}{6}c\Delta x^2(\sigma^2 - 1)k^3$$

Hence

$$u(x, t) = e^{ik(x - ct)} e^{-ik\frac{1}{6}\Delta x^2(\sigma^2 - 1)k^2 t}$$

As phase speed changed the scheme is dispersive

2.4 Extension of upwind scheme

Notes 2.4.1: System of hyperbolic PDE

System of hyperbolic PDE

$$\begin{cases} u_t + \frac{\partial}{\partial x}(f(u, v)) = 0 \\ v_t + \frac{\partial}{\partial x}(g(u, v)) = 0 \end{cases} \iff \begin{pmatrix} u \\ v \end{pmatrix}_t + \begin{pmatrix} f_u & f_v \\ g_u & g_v \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}_x = 0$$

We can generalize to n dimensional and

$$\frac{\partial U}{\partial t} + A(U) \frac{\partial U}{\partial x} = 0$$

If $A(U)$ has real eigenvalue and full set of eigenvector then it is hyperbolic

If $A(U)$ independent of U then linear hyperbolic system

As A is real diagonalizable then

$$A = R^{-1}\Lambda R, \Lambda = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

Then

$$U_t + R^{-1}\Lambda R U_x = 0 \Rightarrow (RU)_t + \Lambda(RU)_x = 0$$

Put $r = RU$:

$$r_t + \Lambda r_x = 0 \Rightarrow \begin{cases} r_{1t} + \lambda_1 r_{1x} = 0 \\ \vdots \\ r_{nt} + \lambda_n r_{nx} = 0 \end{cases}$$

We use upwind method

$$\Lambda = \Lambda_+ + \Lambda_-, \Lambda_+ = \begin{cases} \lambda_i & \lambda_i > 0 \\ 0 & \text{otherwise} \end{cases}, \Lambda_- = \begin{cases} \lambda_i & \lambda_i < 0 \\ 0 & \text{otherwise} \end{cases}$$

The

$$U_t + R^{-1}\Lambda_+ R \frac{U(x_j) - U(x_{j-1}))}{\Delta x} + R\Lambda_- R^{-1} \frac{U(x_{j+1}) - U(x_j)}{\Delta x} = 0$$

Chapter 3

Basics of finite volume method

3.1 Finite volume

Definition 3.1.1: Weak solution

u is a weak solution to

$$u_t + (f(u))_x = 0$$

if for any test function ϕ compactly supported and continuously differentiable we have

$$\int_0^\infty \int_{-\infty}^\infty (u\phi_t + f(u)\phi_x) dx dt = - \int_{-\infty}^\infty \phi(x, 0) u(x, 0) dx$$

Notes 3.1.1: Weak solution

For any compactly supported $\phi(x, t)$ ($\phi(x, t) = 0$ if $|x| > M_x$ or $t > M_t$) continuously differentiable

$$(\phi u)_x = \phi_x u + u_x \phi \Rightarrow u_x \phi = (\phi u)_x - \phi_x u$$

Now to equation

$$u_t + (f(u))_x = 0$$

We have

$$\begin{aligned} \int_0^\infty \int_{-\infty}^\infty (u_t \phi + (f(u))_x \phi) dx dt &= \int_0^\infty \int_{-\infty}^\infty ((\phi u)_t - \phi_t u + (f(u))_x \phi) dx dt \\ &= \int_{-\infty}^\infty \phi(x, \infty) u(x, \infty) - \int_{-\infty}^\infty \phi(x, 0) u(x, 0) \\ &\quad - \int_0^\infty \int_{-\infty}^\infty \phi_t u dx dt + \int_0^\infty [f(u)\phi]_{-\infty}^\infty dt - \int_0^\infty \int_{-\infty}^\infty f(u) \phi_x dx dt \\ &= - \int_{-\infty}^\infty \phi(x, 0) u(x, 0) - \int_0^\infty \int_{-\infty}^\infty \phi_t u dx dt - \int_0^\infty \int_{-\infty}^\infty f(u) \phi_x dx dt = 0 \\ \Rightarrow \int_0^\infty \int_{-\infty}^\infty (u\phi_t + f(u)\phi_x) dx dt &= - \int_{-\infty}^\infty \phi(x, 0) u(x, 0) dx \end{aligned}$$

Notes 3.1.2: Finite Volume method

$$\frac{\partial u}{\partial x} + \frac{\partial f(u)}{\partial x} = 0$$

We integrate over $[x_{i-1/2}, x_{i+1/2}]$

$$\frac{\partial}{\partial t} \int_{x_{i-1/2}}^{x_{i+1/2}} u dx + f(u_{i+1/2}) - f(u_{i-1/2}) = 0$$

Integrate over $[t_n, t_{n+1}]$:

$$\int_{x_{i-1/2}}^{x_{i+1/2}} (u^{n+1} - u^n) dx + \int_{t_n}^{t_{n+1}} (f_{i+1/2} - f_{i-1/2}) dt = 0$$

Define cell average

$$\bar{u}_i = \frac{1}{\Delta x} \int_{x_{i-1/2}}^{x_{i+1/2}} u dx, F_{i+1/2} = \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} f_{i+1/2} dt$$

Then we have

$$\bar{u}_i^{n+1} - \bar{u}_i^n + \frac{\Delta t}{\Delta x} (F_{i+1/2} - F_{i-1/2}) = 0$$

Now we approximate flux

$$\widehat{F_{i+1/2}} = \widehat{F}(\bar{u}_i, \bar{u}_{i+1})$$

to approximate $F_{i+1/2}$

Example 3.1.1: Numerical Flux

(1) Central flux:

$$F(u, v) = F\left(\frac{u+v}{2}\right)$$

(2) Lax-Wendroff flux:

$$F(u, v) = \frac{1}{2}(f(u) + f(v)) + \frac{1}{2} \frac{\Delta t}{\Delta x} \frac{\Delta t}{\Delta u} \frac{1}{2} (u+v)(f(u) + f(v))$$

3.2 Godunov scheme and Riemann problem (Lec 9 not finished)

Notes 3.2.1: Godunov scheme

Logic flow:

$$\frac{\bar{u}_i^{n+1} - \bar{u}_i^n}{\Delta t} = F_{i+1/2}(\bar{u}_{i+1}^n, \bar{u}_i^n) - F_{i-1/2}(\bar{u}_i^n, \bar{u}_{i-1}^n) = 0$$

Best flux: exact solution:

$$F_{i+1/2} = \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} u(x_{i+1/2}) dt$$

Second best: Exact solution to $u_t + \partial_x f(u) = 0$ with IC $u(x, t_n) = \bar{u}_i^n$ if $x \in I_i = [x_{i-1/2}, x_{i+1/2}]$
It is good enough to solve

$$u_t + \partial_x f(u) = 0$$

with

$$u(x) = \begin{cases} \bar{u}_i & x < x_{i+1/2} \\ \bar{u}_{i+1} & x > x_{i+1/2} \end{cases}$$

from t_n to t_{n+1} to approximate $F_{i+1/2}$

Riemann problem:

$$u_t + \partial_x f(u) = 0 \text{ with } u(x, t_n) = \begin{cases} \bar{u}_i & x < x_{i+1/2} \\ \bar{u}_{i+1} & x > x_{i+1/2} \end{cases}$$

from t_n to t_{n+1}

Example 3.2.1: Godunov scheme

$$u_t + cu_x = 0, c > 0$$

$$u(x, 0) = \begin{cases} \bar{u}_i^n & x < 0 \\ \bar{u}_{i+1}^n & x > 0 \end{cases}$$

Then

$$u(x, t) = \begin{cases} \overline{u_i^n} & x - ct < 0 \\ \overline{u_{i+1}^n} & x - ct > 0 \end{cases}$$

If $c > 0$ then $\widehat{cu_{i+1/2}} = u(0, \Delta t) = \overline{u_i^n}$

$$\begin{aligned} \overline{u_i^{n+1}} - \overline{u_i^{n+1}} + \frac{\Delta t}{\Delta x} (\widehat{cu_{i+1/2}} - \widehat{cu_{i-1/2}}) &= 0 \\ \Rightarrow \frac{\overline{u_i^{n+1}} - \overline{u_i^n}}{\Delta t} + c \frac{\overline{u_i^n} - \overline{u_{i-1}^n}}{\Delta x} &= 0 \end{aligned}$$

Which is upwind fd

3.3 Wave equation and pollution effects

Notes 3.3.1: Wave equation

Wave equation

$$\begin{aligned} u_{tt} &= c^2 u_{xx} \\ u(x, 0) &= f(x) \\ u_t(x, 0) &= g(x) \end{aligned}$$

Notes 3.3.2: Yee Scheme

Let $v = u_t, w = cu_x$ then

$$\begin{cases} v_t = cw_x \\ w_t = cv_x \end{cases} \iff u_{tt} = c^2 u_{xx}$$

Then Yee scheme:

$$\begin{aligned} \frac{v_j^{n+1} - v_j^n}{\Delta t} &= c \frac{w_{j+1/2}^{n+1/2} - w_{j-1/2}^{n+1/2}}{\Delta x} \\ \frac{w_{j+1/2}^{n+1/2} - w_{j+1/2}^{n-1/2}}{\Delta x} &= c \frac{v_{j+1}^n - v_j^n}{\Delta x} \end{aligned}$$

Truncation error: $O(\Delta x^2 + \Delta t^2)$

Initialization:

$$\begin{aligned} v_j^0 &= v(x_j, 0) = u_t(x_j, 0) = g(x_j) \\ w_{j+1/2}^{1/2} &= w_{j+1/2}^0 + \frac{1}{2} \Delta t w_{j+1/2, t}^0 + O(\Delta t^2) \\ &= (cu_x)_{j+1/2}^0 + \frac{1}{2} \Delta t (cu_x)_t^0 \\ &= (cf_x)_{j+1/2}^0 + \frac{1}{2} \Delta t (cg_x)_{j+1/2}^0 + O(\Delta t^2) \end{aligned}$$

Put $v_j^n + \lambda^n e^{ikx_j} v, w_{j+1/2}^{n+1/2} = \lambda^{n+1/2} e^{ikx_{j+1/2}} w$

Then

$$\begin{pmatrix} \lambda - 1 & -2i\lambda^{\frac{1}{2}} r \sin \frac{k\Delta x}{2} \\ -2ir \sin \frac{k\Delta x}{2} & \lambda - 1 \end{pmatrix} \begin{pmatrix} v \\ w \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

We want non trivial solution hence

$$\lambda^2 - 2 \left(1 - 2r^2 \sin^2 \frac{k\Delta x}{2} \right) \lambda + 1 = 0$$

Hence we have $\lambda_1 \lambda_2 = 1$

If distinct real root then must have one of $\lambda_1, \lambda_2 > 1$

If same root then $|\lambda| = 1$

If complex root that $\lambda_1 = \overline{\lambda_2}$, hence $\lambda_1 = e^{i\theta}, \lambda_2 = e^{-i\theta}, |\lambda_1| = |\lambda_2| = 1$

Hence scheme stable when $\Delta \leq 0 \iff r = \frac{c\Delta t}{\Delta x} \leq 1$

Notes 3.3.3: Error tolerance

To reach a fixed error tolerance, a p th order FD need $O(k^{\frac{p+1}{p}})$ grid points per direction
Hence for d -dim total grid points $O(k^{d(1+\frac{1}{p})})$

Example 3.3.1: Error tolerance

$$u(x) = e^{ikx} \Rightarrow u_{xx} = -k^2 e^{ikx}$$

Central difference:

$$\begin{aligned} \frac{u_{j+1} - 2u_j + u_{j-1}}{\Delta x^2} &= \frac{-4 \sin^2 \frac{k\Delta x}{2}}{\Delta x^2} e^{ikx} \\ &= -\frac{\sin^2 \frac{k\Delta x}{2}}{(\frac{\Delta x}{2})^2} e^{ikx} \\ \left| k - \frac{\sin^2 \frac{k\Delta x}{2}}{\frac{\Delta x}{2}} \right| &= \left| k - \frac{\frac{k^3 \Delta x^3}{6 \cdot 8}}{\frac{\Delta x}{2}} + O(\Delta x^4) - k \right| \\ &= O(k^3 \Delta x^2) \end{aligned}$$

Then

$$k^3 \Delta x^2 < \varepsilon \Rightarrow \Delta x \sim O(k^{-\frac{3}{2}})$$

Now number of grids $\sim O(1/\Delta x) = O(k^{\frac{3}{2}})$

3.4 FD for elliptic equation and general geometry (Lec 11 not finished)

Notes 3.4.1: elliptic equation

elliptic equation:

$$-\Delta u = f \text{ in } \Omega$$

$$u = g \text{ on } \Omega$$

Central difference:

$$L_h u_{ij} = -\left(\frac{u_{i+1,j} - 2u_{ij} + u_{i-1,j}}{\Delta x^2} + \frac{u_{i,j+1} - 2u_{ij} + u_{i,j-1}}{\Delta y^2} \right) = f_{ij}$$

Matrix vector form: $AU = B$

$$A = \begin{pmatrix} B & -\frac{1}{\Delta y^2} I & & & \\ -\frac{1}{\Delta y^2} I & B & -\frac{1}{\Delta y^2} I & & \\ \ddots & \ddots & \ddots & \ddots & \\ & -\frac{1}{\Delta y^2} I & B & -\frac{1}{\Delta y^2} I & \\ & & -\frac{1}{\Delta y^2} I & B \end{pmatrix}, U = \begin{pmatrix} u_{11} \\ \vdots \\ u_{N_x-1,1} \\ u_{1,2} \\ \vdots \\ u_{N_x-1,2} \\ \vdots \\ u_{N_x-1,N_y-1} \end{pmatrix}$$

$$B = \begin{pmatrix} \frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} & -\frac{1}{\Delta x^2} & & & & \\ -\frac{1}{\Delta x^2} & \frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} & -\frac{1}{\Delta x^2} & & & \\ & & \ddots & \ddots & & \\ & & & -\frac{1}{\Delta x^2} & \frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} & -\frac{1}{\Delta x^2} \\ & & & & -\frac{1}{\Delta x^2} & \frac{2}{\Delta x^2} + \frac{2}{\Delta y^2} \end{pmatrix}$$

Theorem 3.4.1: Max/Min Principle*(1) Continuous level:*Assume $u \in C^2(\overline{\Omega})$ and $-\Delta u = f \leq 0$ in Ω then

$$\min_{x \in \partial\Omega} u \leq \min_{x \in \Omega} u \leq \max_{x \in \Omega} u \leq \max_{x \in \partial\Omega} u$$

*(2) Discret level:*If $L_h u_{ij} = f_{ij} \leq 0$ or ≥ 0 in Ω , the max/min of u_{ij} is obtained at grid points in boundary

Proof.

□

Theorem 3.4.2: Well-posednessIf $U^{(1)}$ and $U^{(2)}$ both numerical solution to $-L_h U = f$ with same boundary condition, then $U_1 = U_2$

Proof.

□

Notes 3.4.2: General geometry

FD approximation:

$$u_{xx}(x_0, y) = \frac{u_1 - 2u_0 + u_{-1}}{\Delta x^2}$$

Boundary condition:

$$u_\eta = g_\Gamma$$

Skip sin future joi write

Chapter 4

Basics of finite element

4.1 Variational form and FEM

Notes 4.1.1: Variational form

Variational form:

$$\begin{aligned} -u_{xx} &= f(x) \quad x \in [0, 1] \\ u(0) &= u(1) = 0 \end{aligned}$$

Test space:

$$H_0^1([0, 1]) = \{v : v, \partial_x v \in L^2[0, 1], v(0) = v(1) = 0\}$$

Where

$$L^2([0, 1]) = \left\{ s : \left| \int_0^1 w^2 dx \right| < \infty \right\}$$

Now

$$\begin{aligned} - \int_0^1 u_{xx} v dx &= \int_0^1 f v dx \\ -u_x(1)v(1) + u_x(0)v(0) + \int_0^1 u_x v_x dx &= \int_0^1 f v dx \\ \int_0^1 u_x v_x dx &= \int_0^1 f v dx \end{aligned}$$

Theorem 4.1.1: equivalence between strong and Variational form

If f and u_{xx} continuous, then

$$-u_{xx} = f \iff \int_0^1 u_x v_x dx = \int_0^1 f v dx \quad \forall v \in H_0^1([0, 1])$$

So Variational problem: seek $u \in H_0^1([0, 1])$ such that

$$(u_x, v_x) = (f, v) \quad \forall v \in H_0^1([0, 1])$$

Where

$$(f, g) = \int_0^1 f g dx$$

Proof. $\Rightarrow$: done

$\Leftarrow$:

$$\int_0^1 u' v' dx = \int_0^1 f v dx \quad \forall v \in H_0^1([0, 1])$$

$$-\int_0^1 u'' v dx + u' v|_0^1 = \int_0^1 f v dx$$

$$\int_0^1 (u'' + f) v = 0$$

If $w = u'' + f \neq 0$, then w is continuous as u'' and f are continuous

Then WLPG there exists $[x_0, x_1] \subseteq [0, 1]$ such that $w > 0$, otherwise consider $-w$

Hence there exists $w_{\min}$ such that $w \geq w_{\min}$ on $[x_0, x_1]$

Then put

$$v = \begin{cases} (x - x_0)^2 (x - x_1)^2 & x \in [x_0, x_1] \\ 0 & \text{otherwise} \end{cases}$$

Then $v \in H_0^1([0, 1])$

Hence

$$\begin{aligned} \int_0^1 (u'' + f) v dx &= \int_0^1 w v dx \\ &= \int_{x_0}^{x_1} w v dx \\ &\geq w_{\min} \int_{x_0}^{x_1} (x - x_0)^2 (x - x_1)^2 > 0 \end{aligned}$$

Contradict to

$$\int_0^1 w v dx = 0 \quad \forall v \in H_0^1([0, 1])$$

□

Notes 4.1.2: Basis functions and discrete problem

Given mesh partition $[0, 1]$, $I_j = [x_{j-1}, x_j]$, $\Delta x_j = x_j - x_{j-1}$ (can be nonuniform) for $j = 0, \dots, N$

$W_{N-1} = \{W : (1) \text{ continuous on } (0, 1), (2) W(x)|_{I_j} \text{ is linear function}, (3) W(0) = W(1) = 0\}$

Basis function for W_N : Hat function

$$\phi_j(x) = \begin{cases} \frac{x - x_{j-1}}{x_j - x_{j-1}} & x \in [x_{j-1}, x_j] \\ \frac{x - x_{j+1}}{x_j - x_{j+1}} & x \in [x_j, x_{j+1}] \\ 0 & \text{otherwise} \end{cases}$$

$$\phi_0(x) = \begin{cases} \frac{-x - x_1}{x_1 - x_0} & x \in [x_0, x_1] \\ 0 & \text{otherwise} \end{cases}$$

$$\phi_{N+1}(x) = \begin{cases} \frac{x - x_N}{x_{N+1} - x_N} & x \in [x_N, x_{N+1}] \\ 0 & \text{otherwise} \end{cases}$$

Now we want to find $u_h = \sum_{j=1}^n u_j \phi_j(x)$ i.e. look for $(u_1, \dots, u_N)$

$$\begin{aligned} (u'_h, \phi'_j) &= (f_h, \phi_j) \\ \left(\sum_{k=1}^N u_k \phi'_k, \phi_j \right) &= (f_h, \phi_j) \\ \sum_{k=1}^N (\phi'_j, \phi'_k) u_k &= (f, \phi_j) \\ SU &= \begin{pmatrix} (\phi'_1, \phi'_1) & \cdots & (\phi'_1, \phi'_N) \\ \vdots & \ddots & \vdots \\ (\phi'_N, \phi'_1) & \cdots & (\phi'_N, \phi'_N) \end{pmatrix} \begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix} = \begin{pmatrix} (f, \phi_1) \\ \vdots \\ (f, \phi_N) \end{pmatrix} = F \end{aligned}$$

Now

$$s_{jk} = (\phi'_j, \phi'_k) = \begin{cases} \frac{-1}{\Delta x_j} & k = j-1 \\ \frac{1}{\Delta x_j} + \frac{1}{\Delta x_{j+1}} & k = j \\ \frac{-1}{\Delta x_{j+1}} & k = j+1 \\ 0 & \text{otherwise} \end{cases}, (\phi'_0, \phi'_j) = \begin{cases} \frac{-1}{x_1 - x_0} & j = 1 \\ 0 & j \neq 1 \end{cases}, (\phi'_{N+1}, \phi'_j) = \begin{cases} \frac{-1}{x_{N+1} - x_N} & j = N \\ 0 & j \neq N \end{cases}$$

And

$$F_j = \int_0^1 f(x) \phi_j(x) dx = \int_{x_{j-1}}^{x_{j+1}} f(x) \phi_j(x) dx = \frac{1}{2} (\Delta x_{j+1} f(x_{j+1/2}) + \Delta x_j f(x_{j-1/2}))$$

Or we can write

$$f_I(x) = \sum_{j=0}^{N+1} f(x_j) \phi_j(x)$$

And hence

$$F_j = \mathbb{I} \begin{pmatrix} f(x_0) \\ f(x_1) \\ \vdots \\ f(x_{N+1}) \end{pmatrix}, \mathbb{I} = \begin{cases} \frac{1}{6} \Delta x_i & k = j-1 \\ \frac{1}{3} (\Delta x_j + \Delta x_{j+1}) & k = j \\ \frac{1}{6} \Delta x_{j+1} & k = j+1 \\ 0 & \text{otherwise} \end{cases}$$

4.2 Boundary conditions and high order FEM

Notes 4.2.1: FEM for BC

STEP 1: Variational form to design suitable trial and test spaces

STEP 2: Discretization: Expand u_h as linear combination of basis function, then we sub into Variational problem

Example 4.2.1: Boundary conditions

(1) Dirichlet BC:

$$\begin{cases} -u_{xx} = f(x) & x \in [0, 1] \\ u(0) = \alpha, u(1) = \beta \end{cases}$$

Variational form:

$$\begin{aligned} - \int_0^1 u_{xx} v dx &= \int_0^1 f v dx \\ \int_0^1 u_x v_x dx - u'(1)v(1) + u'(0)v(0) &= \int_0^1 f v dx \end{aligned}$$

As we no info for $u'(1)$ and $u'(0)$, we kill them by forcing $v(0) = v(1) = 0$

Hence the problem:

Look for

$$u \in W(\alpha, \beta) = \{u \in H^1(0, 1), u(0) = \alpha, u(1) = \beta\}$$

such that

$$\forall v \in H_0^1(0, 1) = \{v \in H^1(0, 1), v(0) = v(1) = 0\}$$

We have

$$(u', v') = (f, v)$$

Discrete problem:

$$u_h \in W_h(\alpha, \beta) = \{(1) \text{ continuous on } (0, 1), (2) u_h|_{I_i} \text{ is linear}, (3) u_h(0) = \alpha, u_h(1) = \beta\}$$

$$u_h = \sum_{k=1}^N u_k \phi_k(x) + \alpha \phi_0(x) + \beta \phi_{N+1}(x)$$

Then

$$\sum_{k=1}^N (\phi_j, \phi'_k) u_k + \alpha (\phi'_j, \phi'_0) + \beta (\phi'_j, \phi'_{N+1}) = F_j$$

Hence

$$SU = F + \begin{pmatrix} \frac{-\alpha}{\Delta x_1} \\ 0 \\ \vdots \\ 0 \\ -\beta \\ \frac{\beta}{\Delta x_{N+1}} \end{pmatrix}$$

(2) Neumann BC:

$$\begin{cases} -u_{xx} = f \\ u(0) = \alpha, u_x(1) = \beta \end{cases}$$

Variational form:

$$\int_0^1 u_x v_x dx - u'(1)v(1) + u'(0)v(0) = \int_0^1 f v dx$$

We know $u_x(1) = \beta$ but no info about $u_x(0)$

Hence we keep $u_x(1)$ but kill $u_x(0)$

Hence the problem:

Find

$$u \in \{H^1(0,1), u(0) = \alpha\}$$

such that for any

$$v \in \{H^1(0,1), u(0) = 0\}$$

such that

$$(u', v') = (f, v) + \beta v(1)$$

Discrete form:

Find

$$u_h \in \{(1) u_h \text{ continuous in } [0,1], (2) u_h|_{I_i} \text{ is linear}, (3) u_h(0) = \alpha\}$$

such that for all

$$v_h \in \{(1) v_h \text{ continuous in } [0,1], (2) v_h|_{I_i} \text{ is linear}, (3) v_h(0) = 0\}$$

i.e.

$$u_h = \sum_{k=1}^{N+1} u_k \phi_k(x) + \alpha \phi_0(x)$$

$$v_h = \sum_{k=1}^{N+1} v_k \phi_k(x)$$

Then

$$\sum_{k=1}^{N+1} (\phi'_j, \phi'_k) u_k = F_j - \alpha (\phi'_j, \phi'_0) + \beta \phi_j(1)$$

$$SU = F + \begin{pmatrix} \alpha \frac{1}{\Delta x} \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

Where

$$S_{N+1,k} = \begin{cases} \frac{-1}{\Delta x_{N+1}} & k = N \\ \frac{1}{\Delta x_{N+1}} & k = N + 1 \\ 0 & \text{otherwise} \end{cases}$$

(3) Robin BC:

$$-u_{xx}(x) = f(x)u(0) = \alpha, u_x(1) + u(1) = \beta$$

We know $u_x(1) = \beta - u(1)$ but no info about $u_x(0)$

Hence we keep $u_x(1)$ but kill $u_x(0)$

Hence the problem:

Find

$$u \in \{H^1(0,1), u(0) = \alpha\}$$

such that for any

$$v \in \{H^1(0,1), u(0) = 0\}$$

such that

$$(u', v') + u(1)v(1) = (f, v) + \beta v(1)$$

Discrete form:

Find

$$u_h \in \{(1) u_h \text{ continuous in } [0,1], (2) u_h|_{I_i} \text{ is linear}, (3) u_h(0) = \alpha\}$$

such that for all

$$v_h \in \{(1) v_h \text{ continuous in } [0,1], (2) v_h|_{I_i} \text{ is linear}, (3) v_h(0) = 0\}$$

i.e.

$$u_h = \sum_{k=1}^{N+1} u_k \phi_k(x) + a \phi_0(x)$$

Then

$$\sum_{k=1}^{N+1} (\phi'_j, \phi'_k) u_k + \phi_j(1) \phi_{N+1}(1) u_{N+1} = F_j - \alpha(\phi'_j, \phi'_0) + \beta \phi_j(1)$$

As

$$\phi_{N+1,k} = \begin{cases} \frac{-1}{\Delta x_N} & k = N \\ \frac{1}{\Delta x_N} + 1 & k = N + 1 \\ 0 & \text{otherwise} \end{cases}$$

Hence

$$SU = F + \begin{pmatrix} \frac{\alpha}{\Delta x} \\ 0 \\ \vdots \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

Notes 4.2.2: Higher order accuracy

We let

$$u_h \in W_N^k = \{(1) u_h \text{ continuous in } [0,1], (2) u_h|_{I_i} \in P^k(I_j) (3) u_h(0) = u_h(1) = 0\}$$

Where $P^k(I_j)$ means polynomials whose deg not greater than k

We add more points inside reference element

$$\phi_{j+\frac{l}{k}} \left(x_j + \frac{l}{k} \right) = \begin{cases} 1 & l = k \\ 0 & \text{otherwise} \end{cases}, l = 0, \dots, k$$

*k*th order polynomial gives *k* + 1th order accuracy

Example 4.2.2: Higher order accuracy

Second order: We need $\{\phi_j\}_{j=1}^N$ and $\{\phi_{j+1/2}\}_{j=1}^N$

$$\phi_j(x) = \begin{cases} \frac{2}{\Delta x_j^2}(x - x_{j-1})(x - x_{j-1/2}) & x_{j-1} \leq x \leq x_j \\ \frac{2}{\Delta x_j^2}(x - x_{j+1})(x - x_{j+1/2}) & x_j \leq x \leq x_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

$$\phi_{j+1/2}(x) = \begin{cases} \frac{-4}{\Delta x_j^2}(x - x_j)(x - x_{j+1}) & x_j \leq x \leq x_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

4.3 Building Blocks for FEM in 2D

Notes 4.3.1: 2D elliptic

Problem:

$$\begin{cases} -\Delta u = -(u_{xx} + u_{yy}) = f & x \in \Omega \\ u|_{\partial\Omega} = 0 \end{cases}$$

Notes 4.3.2: Review in multivariable calculus

Some basic definition:

$$\nabla u = \begin{pmatrix} \partial_{x_1} u \\ \vdots \\ \partial_{x_d} u \end{pmatrix}$$

$$\operatorname{div} \begin{pmatrix} v_1 \\ \vdots \\ v_d \end{pmatrix} = \partial_{x_1} v_1 + \cdots + \partial_{x_d} v_d$$

$$\operatorname{div}(vw) = v \operatorname{div}(w) + \nabla v \cdot w \text{ where } v \text{ is scalar and } w \text{ is vector}$$

$$-\Delta u = \operatorname{div}(\nabla u)$$

Gauss theorem

$$\int_{\Omega} \operatorname{div}(v) dx dy = \int_{\partial\Omega} v \cdot n dS$$

Now

$$\begin{aligned} \int_{\Omega} -\Delta u v dx &= - \int_{\Omega} \operatorname{div}(\nabla u) v dx \\ &= - \int_{\Omega} \operatorname{div}(v \nabla u) dx + \int_{\Omega} \nabla u \cdot \nabla v dx \\ &= - \int_{\partial\Omega} v \nabla u \cdots n dS + \int_{\Omega} \nabla u \cdots \nabla v dx \\ &= \int_{\Omega} \nabla u \cdot \nabla v dx = (\nabla u, \nabla v) \end{aligned}$$

Notes 4.3.3: Variational form

Variational form:

We find

$$u \in H_0^1(\Omega) = \{u : u, \partial_{x_1} u, \dots, \partial_{x_d} u \in L^2(\Omega), u|_{\partial\Omega} = 0\}$$

so that

$$v \in H_0^1(\Omega) = \{u : v, \partial_{x_1} v, \dots, \partial_{x_d} v \in L^2(\Omega), v|_{\partial\Omega} = 0\}$$

such that

$$\langle \nabla u, \nabla v \rangle = \langle f, v \rangle$$

Notes 4.3.4: FEM discretization

Mesh: Partition geometry with triangle

Discrete space = $\{u \text{ continuous}, u|_{T_j} \in P^k(T_j), u|_{\partial\Omega} = 0\}$

Basis function: $k = 1$

Basis: $\lambda_i(x, y)$ satisfy:

(1) Linear on triangle

$$(2) \lambda_i(x_i, y_i) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

If

$$\lambda_i(x, y) = A_i x + B_i y + C_i$$

Then

$$\begin{cases} A_1 x_1 + B_1 y_1 + C_1 = 1 = \lambda_1(x_1, y_1) \\ A_1 x_2 + B_1 y_2 + C_1 = 0 = \lambda_1(x_2, y_2) \\ A_1 x_3 + B_1 y_3 + C_1 = 0 = \lambda_1(x_3, y_3) \end{cases} \Rightarrow \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} \begin{pmatrix} A_1 \\ B_1 \\ C_1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Put $G = \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}$ then

$$\lambda_1(x, y) = \frac{1}{\det G} \det \begin{pmatrix} x & y & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} = \frac{\text{Area of triangle with vertex } (x, y), (x_2, y_2), (x_3, y_3)}{\text{Area of triangle with vertex } (x_1, y_1), (x_2, y_2), (x_3, y_3)}$$

$$\lambda_2(x, y) = \frac{1}{\det G} \det \begin{pmatrix} x_1 & y_1 & 1 \\ x & y & 1 \\ x_3 & y_3 & 1 \end{pmatrix} = \frac{\text{Area of triangle with vertex } (x_1, y_1), (x, y), (x_3, y_3)}{\text{Area of triangle with vertex } (x_1, y_1), (x_2, y_2), (x_3, y_3)}$$

$$\lambda_3(x, y) = \frac{1}{\det G} \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x & y & 1 \end{pmatrix} = \frac{\text{Area of triangle with vertex } (x_1, y_1), (x_2, y_2), (x, y)}{\text{Area of triangle with vertex } (x_1, y_1), (x_2, y_2), (x_3, y_3)}$$

We can notice that

$$\lambda_1(x, y) + \lambda_2(x, y) + \lambda_3(x, y) = 1$$

If we consider $\hat{K}$ as triangle with vertex $\hat{A}_1 = (0, 0)$, $\hat{A}_2 = (1, 0)$, $\hat{A}_3 = (0, 1)$

Then basis function

$$\lambda_1 = 1 - x - y, \lambda_2 = x, \lambda_3 = y$$

Now to convert back to general triangle, we need

$$F \begin{pmatrix} \hat{x} \\ \hat{y} \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Condition:

$$F \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}, F \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}, F \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$$

Therefore we have

$$F \begin{pmatrix} \hat{x} \\ \hat{y} \end{pmatrix} = \begin{pmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{pmatrix} \begin{pmatrix} \hat{x} \\ \hat{y} \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

Then

$$\int_K u(x, y) dx dy = \int_{\hat{K}} u(F(\hat{x}, \hat{y})) |\det J_F| d\hat{x} d\hat{y}$$

with

$$J_F = \begin{pmatrix} \frac{\partial x}{\partial \hat{x}} & \frac{\partial x}{\partial \hat{y}} \\ \frac{\partial y}{\partial \hat{x}} & \frac{\partial y}{\partial \hat{y}} \end{pmatrix}$$

Hence

$$\int_K \nabla u \cdot \nabla v dx dy = |\det J_F| \int_{\hat{K}} (\widehat{\nabla u})^T J_F^{-1} (J_F^{-1})^T \widehat{\nabla v} d\hat{x} d\hat{y}$$

For $k = 2$: we let A_{ij} be mid point of A_i and A_j with A_i, A_j be vertex

Then Basis function: 1 at only one node.

Only 1 at A_i : $\lambda_i(2\lambda_i - 1)$

Only 1 at A_{ij} : $4\lambda_i\lambda_h$

4.4 Discontinuous Galerkin method

Notes 4.4.1:

Model:

$$u_t + cu_x = 0 \quad c > 0, x \in [0, 1]$$

$$u(x, 0) = u_0$$

$$u(0, t) = u_L$$

Discrete space:

$$V_h = V_h^k = \{v : v \in P^k(I_j) \text{ i.e. polynomials on } I_j = [x_{j-1/2}, x_{j+1/2}] \text{ where deg at most } k\}$$

$$[u]_{i+1/2} = u_{i+1/2}^+ - u_{i+1/2}^-$$

$$\{u\}_{i+1/2} = \frac{1}{2}(u_{i+1/2}^+ + u_{i+1/2}^-)$$

Variational form:

$$\int_{I_j} u_{h,t} v_h dx - \int_{I_j} cu_h \partial_x v_h dx + \widehat{cu_{j+1/2}} v_{j+1/2}^- - \widehat{cu_{j-1/2}} v_{j-1/2}^+ = 0$$

where numerical flux

$$\widehat{cu_{j+1/2}} = \begin{cases} cu_{h,j+1/2}^- & c > 0 \\ cu_{h,j-1/2}^+ & c < 0 \end{cases} = c\{u\}_{j+1/2} - \frac{|c|}{2}[u]_{j+1/2}$$

General hyperbolic:

$$u_t + \partial_x(f(u)) = 0$$

Then

$$\int_{I_j} (u_h)_t v_h dx - \int_{I_j} f(u_h) \partial_x v_h dx + \widehat{f_{j+1/2}} v_{j+1/2}^- - \widehat{f_{j-1/2}} v_{j-1/2}^+ = 0$$

where

$$\widehat{f_{j+1/2}} = \hat{f}(u_{j+1/2}^-, u_{j+1/2}^+)$$

(1) Upwind flux:

$$\hat{f}(u^-, u^+) = \begin{cases} f(u^-) & f'(u) > 0 \\ f(u^+) & f'(u) \leq 0 \end{cases}$$

(2) Lax-Friedrichs Flux:

$$\hat{f}(u^-, u^+) = \frac{1}{2}(f(u^-) + f(u^+)) - \max_u |f'(u)|(u^+ - u^-)$$

(3) Godunov flux:

$$\hat{f}(u^-, u^+) = \begin{cases} \min_{u^- \leq u \leq u^+} f(u) & u^- \leq u^+ \\ \max_{u^+ \leq u \leq u^-} f(u) & u^+ \leq u^- \end{cases}$$

Local conservation:

If we pick $v_h|_{I_i} = 1$ then

$$\frac{\partial}{\partial t} \int_{I_j} u_h dx + \hat{f}_{j+1/2} - \hat{f}_{j-1/2} = 0$$

Sum up then

$$\frac{\partial}{\partial t} \int_a^b u_h dx + \hat{f}_{N+1/2} - \hat{f}_{1/2} = 0$$

Implementaion: Basis

$$\xi = \frac{x - x_j}{\frac{1}{2}(x_{j+1/2} - x_{j-1/2})}$$

Legendre polynomials

$$L_0 = 1, L_1 = \xi, L_{k+1} = \frac{2k+1}{k+1} \xi L_k(\xi) - \frac{k}{k+1} L_{k-1}(\xi)$$

It is orthogonal as

$$\int_{-1}^1 L_m(\xi) L_n(\xi) d\xi = \frac{2}{2n+1} \delta_{mn}$$

Lagrange polynomials

$$L_k^{(K)} = \prod_{j=1, j \neq k}^K \frac{\xi - \xi_j}{\xi_k - \xi_j}$$

Then

$$u_h(\xi) = \sum_k u_h(\xi_k) L_k^{(K)}(\xi)$$

Time discretization:

$$MU_t = L(U)$$

(1) Forward Euler:

$$U^{n+1} = U^n + \Delta t M^{-1} L(U^n)$$

(2) SSP-RK2:

$$\begin{aligned} U^{(1)} &= U^n + \Delta t M^{-1} L(U^n) \\ U^{n+1} &= \frac{1}{2} U^n + \frac{1}{2} (U^{(1)} + \Delta t M^{-1} L(U^{(1)})) \end{aligned}$$

(3) SSP-RK3:

$$\begin{aligned} U^{(1)} &= U^n + \Delta t M^{-1} L(U^n) \\ U^{(2)} &= \frac{3}{4} U^n + \frac{1}{4} (U^{(1)} + \Delta t M^{-1} L(U^{(1)})) \\ U^{n+1} &= \frac{1}{3} U^{(1)} + \frac{2}{3} (U^{(2)} + \Delta t M^{-1} L(U^{(2)})) \end{aligned}$$

Stability: $\max_U |f'(u)| \frac{\Delta t}{\Delta x} \leq CFL$ then stable

4.5 Some theoretical aspects of FEM

Definition 4.5.1: Norm, inner product

(1) Norm: $\|\cdot\| : W \rightarrow \mathbb{R}$

Such that

(i) *positive definite*: $\forall v \in W, \|v\| \geq 0$ and $\|v\| = 0 \iff v = 0$

(ii) *Homogeneous*: $\forall \lambda \in \mathbb{R}, v \in W : \|\lambda v\| = |\lambda| \|v\|$

(iii) *Triangular inequality*: $\|v_1 + v_2\| \leq \|v_1\| + \|v_2\| \forall v_1, v_2 \in W$

(2) Inner product: $(\cdot, \cdot) : W \times W \rightarrow \mathbb{R}$

(i) $(x, y) = (y, x) \forall x, y \in W$

(ii) $(ax + by, z) = a(x, z) + b(y, z) \forall x, y, z \in W, a, b \in \mathbb{R}$

(iii) $\forall x \in W, (x, x) \geq 0$ and $(x, x) = 0 \iff x = 0$

Notes 4.5.1: Norm, inner product

For space with inner product, we can define a norm

$$\|x\| = \sqrt{(x, x)}$$

Cauchy Schwartz inequality:

$$|(u, v)| \leq \|u\| \|v\|$$

Example 4.5.1: Norm, inner product

(1) Norm: $v \in H_0^1([0, 1]) = \{v \in L^2([0, 1]), \partial_x v \in L^2([0, 1]), v(0) = v(1) = 0\}$

Energy norm

$$\|v\|_E = \sqrt{\int_0^1 (v_x(x))^2 dx}$$

(2) Inner product: $u, v \in H_0^1([0, 1])$:

$$\langle u, v \rangle_E = \int_0^1 u' v' dx = (u', v')$$

Theorem 4.5.1: Cea's Lemma

FEM results best error in discrete space W_h wrt $\|\cdot\|_E$

$$\|u - u_h\|_E = \|e_h\|_E = \inf_{w_h \in W_h} \|u - w_h\|_E$$

Proof.

$$\begin{aligned} \|u - u_h\|_E &= \|e_h\|_E \geq \inf_{w_h \in W_h} \|u - w_h\|_E \\ (u', v') &= (f, v), \forall v \in H_0^1([0, 1]) \\ (u'_h, v'_h) &= (f, v_h), \forall v_h \in W_h \subseteq H_0^1([0, 1]) \\ \Rightarrow (u', v'_h) &= (f, v'_h) \text{ as } v_h \in W_h \\ (u' - u'_h, v'_h) &= (e'_h, v_h) = 0 \\ \|e_h\|_E^2 &= (e'_h, u' - u'_h) \\ &= (e'_h, u' - w'_h + w'_h - u'_h) \\ &= (e'_h, u' - w'_h) + (e'_h, w'_h - u'_h) \\ &= (e'_h, u' - w'_h) + (e'_h, (w_h - u_h)') \\ &= (e'_h, u' - w'_h) \text{ (As } u_h - w_h \in W_h) \\ \|e_h\|_E^2 &\leq \|e'_h\|_{L^2} \|u' - w'_h\|_{L^2} \\ &\leq \|e_h\|_E \|u - w_h\|_E \end{aligned}$$

$$\begin{aligned}
\|e_h\| &\leq \|u - w_h\|_E \\
\|e_h\| &\leq \inf_{w_h \in W_h} \|u - w_h\|_E \\
\Rightarrow \|u - u_h\|_E = \|e_h\|_E &= \inf_{w_h \in W_h} \|u - w_h\|_E
\end{aligned}$$

□

Theorem 4.5.2: Approximation theory

For any $u \in L^2([0, 1])$, define

$$u_I = \sum_{j=1}^N u(x_j) \phi_j(x)$$

where $\phi_j(x)|_{I_j}$ is linear

Then

$$\|u - u_I\|_E \leq ch \|u_{xx}\|_{L^2}$$

Proof. Let

$$\varepsilon(x) = u(x) - u_I(x)$$

Then

$$\|u - u_I\|_E^2 = \sum_{j=1}^N \int_{x_{j-1}}^{x_j} (\varepsilon'(x))^2 dx$$

Now we focus on $I_j = [x_{j-1}, x_j]$

We have $\varepsilon(x_j) = \varepsilon(x_{j-1}) = 0$

Hence by MVT there exists $\xi \in (x_{j-1}, x_j)$ such that

$$\varepsilon'(\xi) = 0$$

And

$$\begin{aligned}
\varepsilon'(x) &= \int_{\xi}^x \varepsilon''(s) ds + \varepsilon'(\xi) \\
&= \int_{\xi}^x \varepsilon''(s) ds \\
|\varepsilon'(x)| &= \left| \int_{\xi}^x \varepsilon''(s) ds \right| \\
&\leq (x - \xi)^{\frac{1}{2}} \left| \int_{\xi}^x \varepsilon''(s) ds \right|^{\frac{1}{2}} \\
&\leq (x - \xi)^{\frac{1}{2}} \left(\int_{x_{j-1}}^{x_j} \varepsilon''(s) ds \right)^{\frac{1}{2}}
\end{aligned}$$

For all $x \in [x_{j-1}, x_j]$

Then

$$\begin{aligned}
\int_{x_{j-1}}^{x_j} |\varepsilon'(x)|^2 dx &\leq \int_{x_{j-1}}^{x_j} \varepsilon''(s) ds \int_{x_{j-1}}^{x_j} (x - \xi) dx \\
&= \frac{1}{2} ((x_j - \xi)^2 - (x_{j-1} - \xi)^2) \int_{x_{j-1}}^{x_j} \varepsilon''(s) ds \\
&\leq (x_j - x_i)^2 \int_{x_{j-1}}^{x_j} \varepsilon''(s) ds
\end{aligned}$$

Now

$$\begin{aligned}
\varepsilon''(s) &= u''(x) - u_I''(x) \\
&= u''(x)
\end{aligned}$$

as u_I is piecewise linear

Hence

$$\begin{aligned}
 \int_{x_{j-1}}^{x_j} |\varepsilon'(x)|^2 dx &\leq (x_j - x_i)^2 \int_{x_{j-1}}^{x_j} \varepsilon''(s) ds \\
 &= (x_j - x_i)^2 \int_{x_{j-1}}^{x_j} u''(s) ds \\
 &\leq h^2 \int_{x_{j-1}}^{x_j} u''(s) ds \\
 \sum_{j=1}^n \int_{x_{j-1}}^{x_j} |\varepsilon'(x)|^2 dx &\leq \sum_{j=1}^n h^2 \int_{x_{j-1}}^{x_j} u''(s) ds \\
 \int_0^1 |\varepsilon'(x)|^2 dx &\leq h^2 \|u_{xx}\|_{L^2}^2 \\
 \|u - u_I\|_E^2 &\leq h^2 \|u_{xx}\|_{L^2}^2 \\
 \|u - u_I\|_E &\leq ch \|u_{xx}\|_{L^2}
 \end{aligned}$$

□

Theorem 4.5.3: Connect $\|\cdot\|_E$ with $\|\cdot\|_{L^2}$

If

$$\|e_h\|_E \leq h \|f\|_{L^2}$$

Then we have

$$\|e_h\|_{L^2} \leq h^2 \|f\|_{L^2}$$

Proof. We consider dual method:

$$\begin{cases} -w''(x) = e_h(x) \\ w(0) = w(1) = 0 \end{cases}$$

Then

$$\begin{aligned}
 \|e_h\|_{L^2}^2 &= (e_h, e_h) \\
 &= -(e_h, w'') \\
 &= (e_h', w') - e_h(1)w'(1) + e_h(0)w'(0) \\
 &= (e_h', w') \\
 &= (e_h', (w - w_I)') + (e_h', w_I') \\
 &= (e_h', (w - w_I)') + e_h(1)w_I'(1) - e_h(0)w_I'(0) - (e_h, w_I'') \\
 &= |(e_h', (w - w_I)')| \\
 &\leq \|e_h'\|_{L^2} \|(w - w_I)'\|_{L^2} \\
 &= \|e_h\|_E \|w - w_I\|_E \\
 &\leq \|e_h\|_E h \|w''\|_L^2 \\
 &\leq \|e_h\|_E h \|e_h\|_L^2 \\
 \|e_h\|_{L^2} &\leq h \|e_h\|_E \\
 &\leq h^2 \|f\|_{L^2}
 \end{aligned}$$

□

Notes 4.5.2: Error estimate

We can generalize to following:

$$h = \max_j |\Delta x_j|$$

consider

$$\begin{cases} -u_{xx} = f \\ u(0) = u(1) = 0 \end{cases}$$

In FEM we seek

$$u_h \in \{u_h : u_h \in H_0^1([0, 1]), u_h|_{I_j} \in P^k, u_h \text{ continuous}\}$$

If exact solution $u \in L^2[0, 1]$, then error $e_h = u - u_h$ satisfy

$$\|e_h\|_E \leq ch^k \|f^{(k-1)}\|_{L^2}$$

$$\|e_h\|_{L^2} \leq c^2 h^{k+1} \|f^{(k-1)}\|_{L^2}$$

4.6 More on time discretization

Notes 4.6.1: Runge-Kutta method

We convert the PDE into an ODE:

$$y_t = f(t, y)$$

$$\Rightarrow y(t_{n+1}) - y(t_n) = \int_{t_n}^{t_{n+1}} f(t, y) dt$$

An s -stage RK can be represented by Butcher table: $\begin{array}{c|cccc} c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\ c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\ \vdots & \vdots & \vdots & & \vdots \\ c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\ \hline & b_1 & b_2 & \cdots & b_s \end{array}$ where c_i s are ap-

proximate integral using value at $t_n + c_j \Delta t$

And b_i quadrature weights

Y formulation:

$$\begin{cases} Y^{(i)} = y_n + \Delta t + \sum_{j=1}^s a_{ij} f(t_n + c_j \Delta t, Y^{(j)}) \\ y^{n+1} = y_n + \sum_{i=1}^s b_i f(t_n + c_i \Delta t, Y^{(i)}) \end{cases}$$

$\Leftrightarrow K$ formulation:

$$\begin{cases} K^{(i)} = f \left(t_n + c_i \Delta t, y_n + \Delta t \sum_{j=1}^s a_{ij} K^{(j)} \right) \\ y^{n+1} = y_n + \Delta t \sum_{i=1}^s b_i K^{(i)} \end{cases}$$

Implementation: K -formulation cheaper by avoiding repeated function evaluation at $t_n + c_i \Delta t$

Analysis: Y -formulation make things more straight forward

Example 4.6.1: RK method

(1) Implicit mid-point method $O(\Delta t^2)$

$$y^{n+1} = y_n + \Delta t f \left(t_{n+1/2}, \frac{y^{n+1} + y^n}{2} \right)$$

(2) Two-stage explicit, second order accurate RK:

$$0 \leq \alpha \leq 1 : y_{n+\alpha} = y_n + \alpha \Delta t f(t_n, y_n)$$

$$y_{n+1} = y_n + \left(1 - \frac{1}{2\alpha} \right) \Delta t f(t_n, y_n) + \frac{1}{2\alpha} f(t_n + \alpha \Delta t, y_{n+\alpha})$$

$$\alpha = \frac{1}{2} : \text{Explicit mid-point}$$

$$\alpha = \frac{1}{2} : \text{Heun's method}$$

$$\alpha = \frac{2}{3} : \text{Ralston's method}$$

Notes 4.6.2: Absolute stability

We apply ODE solver to $y' = \lambda y$
Then absolute stable region

$$\{\Delta t = (a + bi)\Delta t \in \mathbb{C} \text{ such that } |y_{n+1}| \leq |y_n|\}$$

For PDE, one can analyse stability by:

STEP 1: Locally linearize it

STEP 2: Put $u(x_j, t) = e^{ikx_j} \hat{u}(t)$ into scheme after linearization

STEP 3: Figure out condition for Δt so that $\lambda \Delta t$ inside stability region

Example 4.6.2: Absolute stability

Forward Euler:

$$y_{n+1} = (1 + \lambda \Delta t) y_n$$

Then

$$|y_{n+1}| \leq |y_n| \Rightarrow |1 + \lambda \Delta t| = |-1 - (\lambda \Delta t)| \leq 1$$

Therefore $\lambda \Delta t \in$ circle centered at -1 with radius 1

Now consider

$$u_{jt} + c \frac{u_{j+1} - u_{j-1}}{2\Delta x} = 0, u_j = e^{ikx_j} \hat{u}$$

Then

$$\hat{u}_t + ic \sin(k\Delta x) \hat{u} = 0 \Rightarrow \lambda = -ic \sin(k\Delta x)$$

Which not in region for all Δx , hence Unconditional unstable

Backward Euler:

$$(1 - \lambda \Delta t) y_{n+1} = y_n$$

Then

$$|y_{n+1}| \leq |y_n| \Rightarrow |1 - \lambda \Delta t| \geq 1$$

Notes 4.6.3: Order reduction of RK

For RK4, if u_{-2} and u_{-1} are obtained by polynomial extrapolation, then we will not see 4th order accuracy

Reason: RK relies on perfect cancellation in Taylor expansion

Fix: Lax-Wendroff flavour boundary treatment by repeatedly using PDE near boundary

Example 4.6.3:

$$\begin{aligned} u_t + u_x &= 0 \\ u(0) &= g(t) \end{aligned}$$

Then

$$\begin{aligned} u_{-1}^n &= u_0^n - \Delta x u_{0x}^n + \frac{1}{2} \Delta x^2 u_{0xx}^n + \dots \\ &= u^n - \Delta x u_{0t}^n + \frac{1}{2} \Delta x^2 u_{0tt}^n + \dots \\ &= u^n - \Delta x g_t(t_n) + \frac{1}{2} \Delta x^2 g_{tt}(t_n) + \dots \end{aligned}$$

Notes 4.6.4: Multistep method

Use $t_n, t_{n-1}, \dots, t_{n-s}$ to construct polynomials approximation to $f(t, y) = Pf(t, y)$ then

$$\int_{t_n}^{t_{n+1}} f(t, y) dt = \int_{t_n}^{t_{n+1}} Pf(t, y) dt$$

Initialization: For k th order accurate scheme, to compute y^k , we need $y^0, \dots, y^{k-1}$

Example 4.6.4: Multistep method

(1) Adam-Bashford:

$$y^n = y^{n-1} + \Delta t \sum_{j=1}^k \beta_j f^{n-j}$$

$$\beta_j = (-1)^{j-1} \sum_{i=j-1}^{k-1} \binom{i}{j-1} r_i$$

$$r_i = (-1)^i \int_0^1 \binom{-s}{i} ds$$

Second order:

$$y^{n+1} = y^n + \frac{\Delta t}{2} (3f^n - f^{n-1})$$

Third order:

$$y^{n+1} = y^n + \frac{\Delta t}{12} (23f^n - 16f^{n-1} + 5f^{n-2})$$

(2) Adam-Moulton:

Second order:

$$y^{n+1} = y^n + \frac{\Delta t}{12} (5f^{n+1} + 8f^n - f^{n-1})$$

Third order:

$$y^{n+1} = y^n + \frac{\Delta t}{24} (9f^{n+1} + 19f^n - 5f^{n-1} + f^{n-2})$$

(3) : BDF

$$\begin{aligned} \text{BDF1: } y^{n+1} &= y^n + \Delta t f(t_{n+1}, y^{n+1}) \\ \text{BDF2: } y^{n+1} &= \frac{4}{3}y^n - \frac{1}{3}y^{n-1} + \frac{2}{3}\Delta t f(t_{n+1}, y^{n+1}) \\ \text{BDF3: } y^{n+1} &= \frac{18}{11}y^n - \frac{4}{11}y^{n-1} + \frac{2}{11}y^{n-2} + \frac{6}{11}\Delta t f(t_{n+1}, y^{n+1}) \end{aligned}$$

To initialize: STEP 1:

$$y^1 = y^0 + \Delta t f^1 / y^0 + \Delta t f^0$$

STEP 2: BDF2 $O(\Delta t^2)$

Notes 4.6.5: Implicit-Explicit(IMEX) or operator splitting

Consider

$$u_t + \partial_x(f(u)) = \varepsilon u_{xx}$$

Then IMEX:

$$\frac{u^{n+1} - u^n}{\Delta t} + \partial_x f(u^n) = \partial_{xx} u^{n+1}$$

Operator splitting

(1) Lie-Splitting:

(a) Advection step:

$$\frac{u^{n,1} - u^n}{\Delta t} + \partial_x(f(u^n)) = 0$$

(b) Diffusion step:

$$\frac{u^{n+1} - u^{n,1}}{\Delta t} = \partial_{xx} u^{n+1}$$

$O(\Delta t)$ error

(2) Strong splitting

(a) Half setp advection:

$$\frac{u^{n,1} - u^n}{\frac{1}{2}\Delta t} + \partial_x(f(u^n)) = 0$$

(b) Full Step diffusion

$$\frac{u^{n,2} - u^{n,1}}{\Delta t} = \partial_{xx} u^{n+1}$$

(c) Half setp advection:

$$\frac{u^{n+1} - u^{n,2}}{\frac{1}{2}\Delta t} + \partial_x(f(u^{n,2})) = 0$$

$O(\Delta t^2)$ error

Chapter 5

Basic linear equation solver

5.1 Basic iterative linear solvers

Notes 5.1.1: Direct solvers

$A = LU$, L and U are upper triangular matrix

$$Ax = b \Rightarrow Ly = b, Ux = y$$

If bandwidth of matrix is b the LU: cost $O(Nb^2)$, memory $O(Nb)$

Notes 5.1.2: Iterative method

$Ax = b$ when A large and sparse

$$Ax = b \iff x = Gx + c \Rightarrow x^{(k+1)} = Gx^{(k)} + c$$

If iteration converge

$$\lim_{k \rightarrow \infty} x^{(k)} = x, \text{ where } Ax = b$$

Theorem 5.1.1: Convergence

$x^{(k+1)} = Gx^{(k)} + c$ converge to $x = Gx + c$ as $k \rightarrow \infty \iff \rho(G) = \max |\lambda| < 1$

Proof.

$$\begin{aligned} x &= Gx + c \\ x^{(k+1)} &= Gx^{(k)} + c \\ e^{(k)} &= x - x^{(k)} \\ e^{(k+1)} &= Ge^{(k)} \\ &= \dots \\ &= G^{k+1}e^{(0)} \\ \rho(G) < 1 &\iff \lim_{k \rightarrow \infty} G^k = 0 \\ &\iff \lim_{k \rightarrow \infty} e^{(k)} = 0 \end{aligned}$$

□

Theorem 5.1.2: Rate of convergence

Error bound:

$$\|e^{(k)}\| \leq \frac{\|G\|^k}{1 - \|G\|} \|x^{(1)} - x^{(0)}\|$$

for $\|G\| < 1$

Proof.

$$\begin{aligned}
 \|e^0\| &= \|x - x^{(0)}\| \|x - x^{(1)} + x^{(1)} - x^{(0)}\| \\
 &= \|x - x^{(1)}\| + \|x^{(1)} - x^{(0)}\| \\
 &= e^{(1)} + \|x^{(1)} - x^{(0)}\| \\
 &\leq \|G\| \|e^{(0)}\| + \|x^{(1)} - x^{(0)}\| \\
 &\leq \frac{\|x^{(1)} - x^{(0)}\|}{1 - \|G\|} \\
 \|e^{(k)}\| &\leq \frac{\|G\|^k}{1 - \|G\|} \|x^{(1)} - x^{(0)}\|
 \end{aligned}$$

□

Example 5.1.1: Iterative method

$A = D - L - U$ where

$$D = \begin{pmatrix} a_{11} & & \\ & \ddots & \\ & & a_{nn} \end{pmatrix}, L = \begin{pmatrix} 0 & & \\ -a_{21} & 0 & \\ \vdots & \ddots & \ddots \\ -a_{n,1} & -a_{n,2} & \cdots & 0 \end{pmatrix}, U = \begin{pmatrix} 0 & -a_{12} & \cdots & -a_{1n} \\ & \ddots & & \vdots \\ & & \ddots & -a_{n,n-1} \\ & & s & 0 \end{pmatrix}$$

Then

$$\begin{aligned}
 Ax = b &\iff (D - L - U)x = b \\
 &\iff Dx = b + (L + U)x \\
 &\iff x = D^{-1}(L + U)x + D^{-1}b
 \end{aligned}$$

(1) *Jacobi iteration:*

$$x^{(k+1)} = D^{-1}(b + Lx^{(k)} + Ux^{(k)})$$

(2) *Gauss-siedel iteration*

$$x^{(k+1)} = (D - L)^{-1}(Ux^{(k)} + b)$$

(3) *Successive over relaxation*

$$x^{(k+1)} = (D - \omega L)^{-1}((1 - \omega)Dx^{(k)} + \omega(b + Ux^{(k)}))$$

Choosing optimal ω , SOR converge faster

(4) *Thomas Algorithm for tridiagonal matrices*

$$\begin{pmatrix} b_1 & c_1 & & \\ a_2 & b_2 & c_2 & \\ \ddots & \ddots & \ddots & \\ & & c_{n-1} & \\ & a_n & b_n & \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_n \end{pmatrix}$$

Then

$$\begin{aligned}
 x_i + c'_i x_{i+1} &= d'_i \\
 x_n &= d'_n
 \end{aligned}$$

where

$$d'_i = \begin{cases} d_i & i = 1 \\ \frac{d_i - a_i d'_{i-1}}{b_i - a_i c'_{i-1}} & i \geq 1 \end{cases}, \quad c'_i = \begin{cases} \frac{c_i}{b_i} & i = 1 \\ \frac{c_i}{b_i - a_i c'_{i-1}} & i \geq 1 \end{cases}$$

5.2 Krylov method and preconditioning

Definition 5.2.1: Krylov subspace

For iterative method, x_0 initial guess, x_k be the result of k th iteration:

$$r_0 = b - Ax_0, r_k = b - Ax_k$$

Then

$$K(r_0, A, m) = \text{Span}\{r_0, \dots, A^{m-1}r_0\}$$

If $m = n$ then

$$K(r_0, A, n) = \mathbb{R}^n$$

Notes 5.2.1: Krylov method

In m th iteration, we seek

$$x_m \in x_0 + K_m(r_0, A, m) = \text{Span}\{x_0 + a_0 r_0 + \dots + a_{m-1} A^{m-1} r_0\} = \text{Span}\{x_0 + q_{m-1}(A) r_0\}$$

to minimize error inside Krylov subspace

Krylov subspace can be implicitly generated as iteration continue

Krylov method aim to use info provided by entire iteration history through this subspace

Notes 5.2.2: General Minimal residual method (GMRES)

Idea: minimize $r_m = b - Ax_m$ in $x_0 + K(r_0, A, m)$

(1) Algorithm:

Compute $r_0 = b - Ax_0, \beta = \|r_0\|, v_1 = \frac{r_0}{\beta}$

for $j=1:m$ **do**

 Compute $w_j = Av_j$ $\triangleright$ Compute Krylov vector to expand Krylov subspace by one dimension

 Build orthonormal bases for Krylov subspace:

for $i=1:j$ **do**

$$H_{ij} = w_j^T v_i$$

$$w_j = w_j - H_{ij} v_i$$

end for

if $H_{j+1,j} = 0$ **then**

break

else

$$H_{j+1,j} = \|w_j\|_2$$

$$v_{j+1} = \frac{w_j}{h_{j+1,j}}$$

end if

Find $x_k = x_0 + V_k y$ to minimize residual $\|Ax_k - b\|$ in $x_0 + K(r_0, A, k)$:

$$\text{Solve } y = \min_{\tilde{y}} \|H\tilde{y} - \beta e_1\| \text{ and } e_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

compute residual $\|Hy - \beta e_1\|$

if Residual small enough **then**

 Break

end if

end for

$$y = \min_{\tilde{y}} \|H\tilde{y} - \beta e_1\|, x_m = x_0 + V_m y_m$$

(2) Key steps:

$$x_m \in K(r_0, A, m) + x_0$$

$$\begin{aligned}
b - Ax_m &= b - A(x_0 + V_m y) \\
&= r_0 - Av_m y \\
&= \beta v_1 - Av_m y \\
&= \beta v_{m+1} e_1 - v_{m+1} H_m y \\
&= (v_{m+1})(\beta e_1 - H_m y)
\end{aligned}$$

Now v_{m+1} has orthonormal column:

$$\begin{aligned}
v_{m+1}^T v_{m+1} &= 1 \\
\|v_{m+1} y\|_2^2 &= \|y\|_2^2 \\
\|b - Ax_m\| &= \|\beta e_1 - H_m y\|
\end{aligned}$$

Hence find $x_m = x_0 + v_m y$ minimize $\|b - Ax_m\| \iff$ finding y minimize $\|\beta e_1 - H_m y\|$

(3) Convergence theory:

If

$$A = X \Lambda X^{-1}, \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$$

Then

$$\|r_n\|_2 \leq \kappa_2(x) \varepsilon^{(n)} \|r_0\|_2$$

where

$$\varepsilon^{(n)} = \min_{p(0)=1} \max_{i=1, \dots, N} |p(\lambda_i)|$$

with p polynomial degree n

Notes 5.2.3: Conjugate Gradient Method(CG)

Only applicable when A is SPD ($A^T = A$ and $x^T A x > 0 \forall x \neq 0$)

If A is SPD then $\|x\|_A = \sqrt{x^T A x}$ defines a norm

Idea: minimize $\|x - x_m\|_A$ in $x_0 + K(r_0, A, m)$

(1) Algorithm:

$$r_0 = b - Ax_0, p_0 = r_0, q_0 = Ar_0$$

for $j = 1 : m$ **do**

$$\alpha_j = \frac{r_j^T r_j}{q_j^T q_j}$$

$$x_{j+1} = x_j + \alpha p_j$$

$$q_j = Ap_j$$

$$r_{j+1} = r_j - \alpha_j q_j$$

$$\beta_j = \frac{r_{j+1}^T r_{j+1}}{r_j^T r_j}$$

$$p_{j+1} = r_{j+1} + \beta_j p_j$$

end for

(2) CG can be derived by minimizing $\|b - Ax_m\|_2$ in $x_0 + K_m(r_0, A, m)$ via replacing Arnoldi iteration with Lacaos iteration using 3 terms recurrence relation specially designed for SPD

(3) Convergence:

If A SPD:

$$\begin{aligned}
\|x - x_n\|_A &\leq 2 \left(\frac{\sqrt{\kappa(A)} - 1}{\sqrt{\kappa(A)} + 1} \right)^n \|x - x_0\|_A \\
\kappa(A) &= \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)}
\end{aligned}$$

Notes 5.2.4: Preconditioner

Error decay: Krylov method: $1 - O\left(\frac{1}{\kappa(A)}\right)$, Jacobi/GS: $1 - O\left(\frac{1}{\kappa(A)}\right)$

$$Ax = b \iff M^{-1}Ax = M^{-1}b \iff AM^{-1}y = b, x = M^{-1}y$$

Hence if AM^{-1} or $M^{-1}A$ has smaller condition number, iterative method will converge faster
Preconditioner:

- (1) close to A^{-1}
- (2) Computationally cheap
- (3) $M^{-1}A$ or AM^{-1} has better condition number

Cases of preconditioning:

(a) Left preconditioning:

Solve

$$M^{-1}Ax = M^{-1}b$$

Krylov subspace:

$$x_0 + \text{Span}\{r_0, M^{-1}Ar_0, \dots, (M^{-1}A)^m r_0\}$$

Left preconditioned GMRES minimize $\|M^{-1}Ax - M^{-1}b\|$ in preconditioned subspace

(b) Right preconditioning:

Solve

$$AM^{-1}y = b \Rightarrow x = M^{-1}y$$

Then

$$y_m \in y_0 + \text{Span}\{r_0, AM^{-1}r_0, \dots, (AM^{-1})^{m-1}r_0\}$$

$$x_m = M^{-1}y_m \in x_0 + \text{Span}\{M^{-1}r_0, M^{-1}AM^{-1}r_0, \dots, M^{-1}(AM^{-1})^{m-1}r_0\}$$

5.3 Basic Introduction to multigrid

Notes 5.3.1:

Jacobi iteration:

$$x_{k+1} = D^{-1}((L + U)x_k + b)$$

$$= Gx_k + g$$

$$x = Gx + g$$

$$\Rightarrow e_{k+1} = Ge_k$$

Eigenvalue and eigenvector:

$$\lambda_n = \cos \frac{n\pi}{N}, u_n = \left(\sin \frac{n\pi}{N}, \dots, \sin \frac{(N-1)n\pi}{N} \right)$$

Relaxed Jacobi iteration

$$x_{k+1} = (1 - \omega)x_k + \omega D^{-1}((L + U)x_k + b)$$

$$= ((1 - \omega)I + \omega G)x_k + g_\omega$$

$$G_\omega = (1 - \omega)I + \omega G$$

$$\lambda_n = 1 - \omega + \omega \cos \frac{n\pi}{N}$$

$$e_{k+1} = \sum_{n=1}^{N-1} e_{k,n} G_\omega u_n$$

$$= \sum_{n=1}^{N-1} e_{k,n} \lambda_n(\omega) u_n$$

$$e_{k+1,n} = e_{k,n} \lambda_n(\omega)$$

$$e_{k,n} = \sum_{n=1}^{N-1} \lambda_n^k e_{0,n} u_n$$

For $0 < \omega < 1$: high frequency error decays faster

Smooth error component on fine grid look more oscillatory on coarse grid

$$u_1 \text{ with } \Delta x = \frac{1}{N} \Rightarrow \text{Decay factor is } \cos \frac{\pi}{N}$$

$$u_1 \text{ with } \Delta x = \frac{2}{N} \Rightarrow \text{Decay factor is } \cos \frac{2\pi}{N}$$

If evaluating error correspond to $\sin(n\pi x)$, then decay factor smaller on coarser mesh

- (1) Fine grid can kill high frequency error more effectively by choosing suitable ω
- (2) Moving from fine grid to coarser grid, we can more effectively eliminate low frequency error decaying slowly on original fine grid (frequency $n < N/2$)
- (3) Highest frequency component on fine grid is low frequency on coarse grid (frequency $n > N/2$)

Notes 5.3.2: Multigrid method

General idea:

- (1) Fine grid for high frequency
- (2) Coarse grid for low frequency

Information transfer between grids

(a) Coarse to Fine:

Interpolation:

$$\Omega^{2h} \rightarrow \Omega^h$$

$$\mathbb{R}^{N/2-1} \rightarrow \mathbb{R}^{N-1}$$

$$I_{2h}^h v^{2h} = v^h$$

$$0 \leq j \leq \frac{N}{2} - 1 : v_{2j}^h = v_j^{2h}$$

$$v_{2j+1}^h = \frac{1}{2} v_j^{2h} + v_{j+1}^{2h}$$

$$v_0^{2h} = v_{N/2}^{2h} = 0$$

Linear Interpolation:

$$I_{2h}^h = \frac{1}{2} \begin{pmatrix} 1 & & & \\ 2 & & & \\ 1 & 1 & & \\ & 2 & & \\ & & 1 & 1 & \ddots \end{pmatrix}$$

(b) Restriction: Fine to Coarse

Full weighting:

$$v_j^{2h} = \frac{1}{4} (v_{2j-1}^h + 2v_{2j}^h + v_{2j+1}^h)$$

Matrix:

$$I_h^{2h} = \frac{1}{4} \begin{pmatrix} 1 & 2 & 1 & & \\ & 1 & 2 & 1 & \\ & & 1 & 2 & 1 \end{pmatrix}$$

Hence

$$I_{2h}^h = 2(I_h^{2h})^T$$

Notes 5.3.3: Two-grid method (MG2)

(1) Relax (apply Jacobi/weighted Jacobi/GS/SOR) v_1 times on $A^h u^h = f$ on Ω^h

(2) Compute fine grid residual

$$r^h = f^h - A^h u^h$$

And restrict to coarse grid

$$r^{2h} = I_h^{2h} r^h$$

(3) SOLve coarse grid problem

$$A^{2h} e^{2h} = r^{2h} \text{ on } \Omega^{2h}$$

Where

$$A^{2h} = I_h^{2h} A_h I_{2h}^h = c(I_{2h}^h)^T A_h I_{2h}^h$$

(4) Interpolate coarse grid error

$$e^h = I_{2h}^h e^{2h}$$

and update

$$u^h = u^h + e^{2h}$$

(5) Relax $A^h u^h = f^h$ on Ω^h by Jacobi/GS/SOR for v_2 times
Repeat until converge

Notes 5.3.4: MG

(1) $A^{2h} e^{2h} = r^{2h}$ can be solved by another two grid procedure:

V-cycle: $\Omega_h \rightarrow \Omega_{2h} \rightarrow \Omega_{4h} \rightarrow \Omega_{2h} \rightarrow \Omega_h$

W-cycle: $\Omega_h \rightarrow \Omega_{2h} \rightarrow \Omega_{4h} \rightarrow \Omega_{2h} \rightarrow \Omega_{4h} \rightarrow \Omega_{2h} \rightarrow \Omega_h$

(2) $O(M)$ cost in MG with V-cycle or W-cycle

Mat vec multiply cost in $O(M)$, hence method converge within fixed number of iteration regardless of grid size

(3) Algebraic MG: MG without grid, coarse grid is algebraically built based on info in A

(4) MG as preconditioner: $M^{-1} \approx A^{-1}$ by apply a few MG

Combine with CG: Implemented by replacing Jacobi/SOR/GS in relaxation step with one CG iteration

5.4 On the nonlinear system solver**Notes 5.4.1: Nonlinear solver**

We need to solve a non linear equation or system:

$$u = G(u) \text{ or } F(u) = 0$$

Notes 5.4.2: Fixed point method

For

$$u = G(u)$$

We can iterate

$$u_{n+1} = G(u_n)$$

Definition 5.4.1: Cauchy sequence, complete

$\{u_n\}_{n=1}^\infty$ is a Cauchy sequence if $\forall \varepsilon > 0, \exists N$ such that $n > N, m > 0$ we have $\|u_{m+n} - u_n\| < \varepsilon$
 U is complete if every cauchy sequence in U converge to a limit

Theorem 5.4.1: Fixed point method

If $G : U \rightarrow U$ and U complete, if G is contraction mapping (there exists $0 < L < 1$ such that $\|G(x) - G(y)\| \leq L\|x - y\|$), then fixed point iteration converge to u that $u = G(u)$

Proof.

$$\begin{aligned}
 \|u_{n+1} - u_n\| &= \|G(u_n) - G(u_{n-1})\| \\
 &\leq L\|u_n - u_{n-1}\| \\
 &\dots \\
 &\leq L^n\|u_1 - u_0\| \\
 \|u_{m+n} - u_n\| &\leq \|u_{m+n} - u_{m+n-1}\| + \dots + \|u_{n+1} - u_n\| \\
 &\leq (L^{m+n} + \dots + L^n)\|u_1 - u_0\| \\
 &\leq \frac{L^n}{1-L}\|u_1 - u_0\| \xrightarrow{n \rightarrow \infty} 0 \text{ as } 0 \leq L < 1
 \end{aligned}$$

Hence sequence is cauchy

Now as U complete, there exists $u \in U$ such that $\lim_{n \rightarrow \infty} u_n = u$

$$\begin{aligned}
 \|u - G(u)\| &\leq \|u - u_n\| + \|G(u) - G(u_n)\| \\
 &= \|u - u_n\| + \|G(u) - G(u_{n-1})\| \\
 &\leq \|u - u_n\| + L\|u - u_{n-1}\| \rightarrow 0
 \end{aligned}$$

□

Notes 5.4.3:

(1) We need G to be contraction:

$$u = 2u, u = 0$$

If $u^{n+1} = 2u^n$ then

$$u^1 = 1, u^2 = 2, \dots, u^n = 2^n$$

(2) Pseudo time matching to solve $F(u) = 0$

Forward Euler:

$$\begin{aligned}
 u_{n+1} &= u_n + \Delta t F(u_n) \\
 &= (I + \Delta t F)(u_n)
 \end{aligned}$$

$$u_{xx} = 0 \Rightarrow (I + \Delta t \partial_{xx}) \rightarrow \text{ev between 0 and 1 for small } \Delta t$$

(3) Gradient descent:

$$u^{n+1} = u^n - r \nabla F(u^n)$$

It can be seen as fixed point iteration solving $-\nabla F(u) = 0$, minimizer of $F(u)$ must satisfy this

Notes 5.4.4: Anderson Acceleration

Given x_0 and $m \geq 1$

for $k = 1$: Maximum of iteration **do**

Set $m_k = \min\{m, k\}$

Set $R_k = (r_{k-m_k}, \dots, r_k)$ where $r_k = g(x_k) - x_k$

Compute $\alpha^{(k)} = (\alpha_0^{(k)}, \dots, \alpha_{m_k}^{(k)})^T$ such that

$$\alpha^{(k)} = \operatorname{argmin}_{\alpha \in \mathbb{R}^{m_k+1}} \|R_k \alpha\|_2$$

with

$$\sum_{j=0}^{m_k} \alpha_j^{(k)} = 1$$

▷ Seek x_{k+1} as linear combination of $G(x_k), \dots, G(x_{k-m_k})$ such that residual is minimized

$$x_{k+1} = \sum_{j=0}^{m_k} \alpha_j^{(k)} g(x_{k-m_k+j})$$

if Residual_ithreshold **then**

Break

end if

end for

If $m = +\infty$ and $F(u) = Au - b$ then Anderson Acceleration is almost GMRES

Notes 5.4.5: Newton's method

Solve $F(u) = 0$

Taylor expansion:

$$0 = F(u) = F(u_k) + J_k(u - u_k) + \text{high order term}$$

$$u \approx u_k - J_k^{-1} F(u_k)$$

$$u_{k+1} = u_k - J_k^{-1} F(u_k)$$

$$J_k = \begin{pmatrix} \nabla^T F_1 \\ \vdots \\ \nabla^T F_n \end{pmatrix} = \begin{pmatrix} \frac{\partial F_1}{\partial u_1} & \cdots & \frac{\partial F_1}{\partial u_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial u_1} & \cdots & \frac{\partial F_n}{\partial u_n} \end{pmatrix}$$

Notes 5.4.6: Jacobi-Free-Newton-Krylov(JFNK)

Apply preconditioned Krylov solver to solve $J_k \delta u^{(k)} = -F(u_k)$

But sometimes $J_k = J(u_k)$ is hard to obtain

So we solve with Krylov method, i.e. we need

$$\delta u_0 + \text{Span}\{r^{(0)}, J_k r^{(0)}, \dots, J_k^{m-1} r^{(0)}\}$$

Hence we need J_k krylov vector v which is

$$J_k \cdot v \approx \frac{F(u^{(k)} + \varepsilon v) - F(u^{(k)} - \varepsilon v)}{2\varepsilon}$$

Preconditioning is essential

Inexact Newton: We can stop inner loop of Krylov solvers early based on accuracy criteria e.g.

$$\|J_k \delta u^{(k)} + F(u_k)\|_2 \leq \theta_k \|F(u_k)\|, \quad 0 < \theta_k < 1$$

As in early stage we far from exact solution hence solve accurately not contribute to overall convergence

Early: Inexact solve (save cost)

Latter: Accurate solve (Accuracy)

Notes 5.4.7: Relative advantage and disadvantage

(1) Newton Krylov

Advantage:

- (i) Faster convergence
- (ii) Scalable on parallel machine

Disadvantage

- (i) Require Jacobi
- (ii) Require good initial guess

(2) JFNK

Advantage:

- (i) Jacobi-Free

(ii) Scalable

Disadvantage

(i) FD approx introduce error

(ii) sensitive to initial guess

(iii) Sometimes slower than Newton

(3) Anderson Acceleration

Advantage:

(i) Jacobi-Free

(ii) Scalable

Disadvantage

(i) Require careful design of fixed point problem

(ii) sometimes less robust than Newton

(4) Fixed point iteration

Advantage:

(i) Simple

Disadvantage

(i) Slow, require careful design